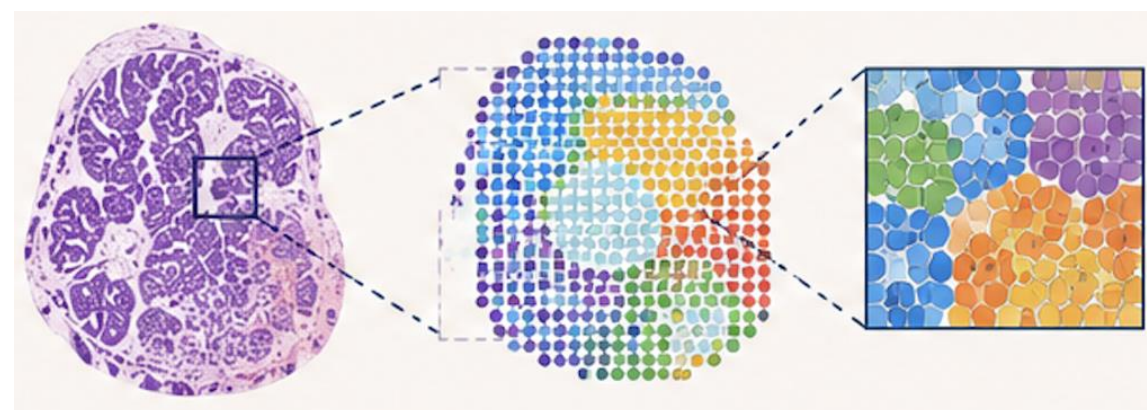


Introduction to spatial transcriptomics—

From molecules to maps: How spatial measurements reveal biology



Arpita Kulkarni, Ph.D.

HBC Spatial Transcriptomics Workshop • October 2026



How this talk is built

Six sections. One throughline: preserve the information your biological question needs.

01 WHY

What do we lose when tissue is dissociated?

02 HOW

How can molecular measurements retain spatial information?

03 RESOLUTION

What does “resolution” actually mean?

04 CHOICE

How do we match a technology to a biological question?

05 INTERPRETATION

How does a measurement become a biological map?

06 HANDOFF

What will you actually do with spatial data?

Central message

- Spatial transcriptomics is not one technology. It is a family of measurement strategies that trade transcriptome breadth, spatial granularity, tissue coverage, and interpretability differently

Practical rule:

- Start with the biological question. Decide what information you cannot afford to lose. Then choose the platform that captures the measurement
- A spatial map is evidence — not biological truth — until validated

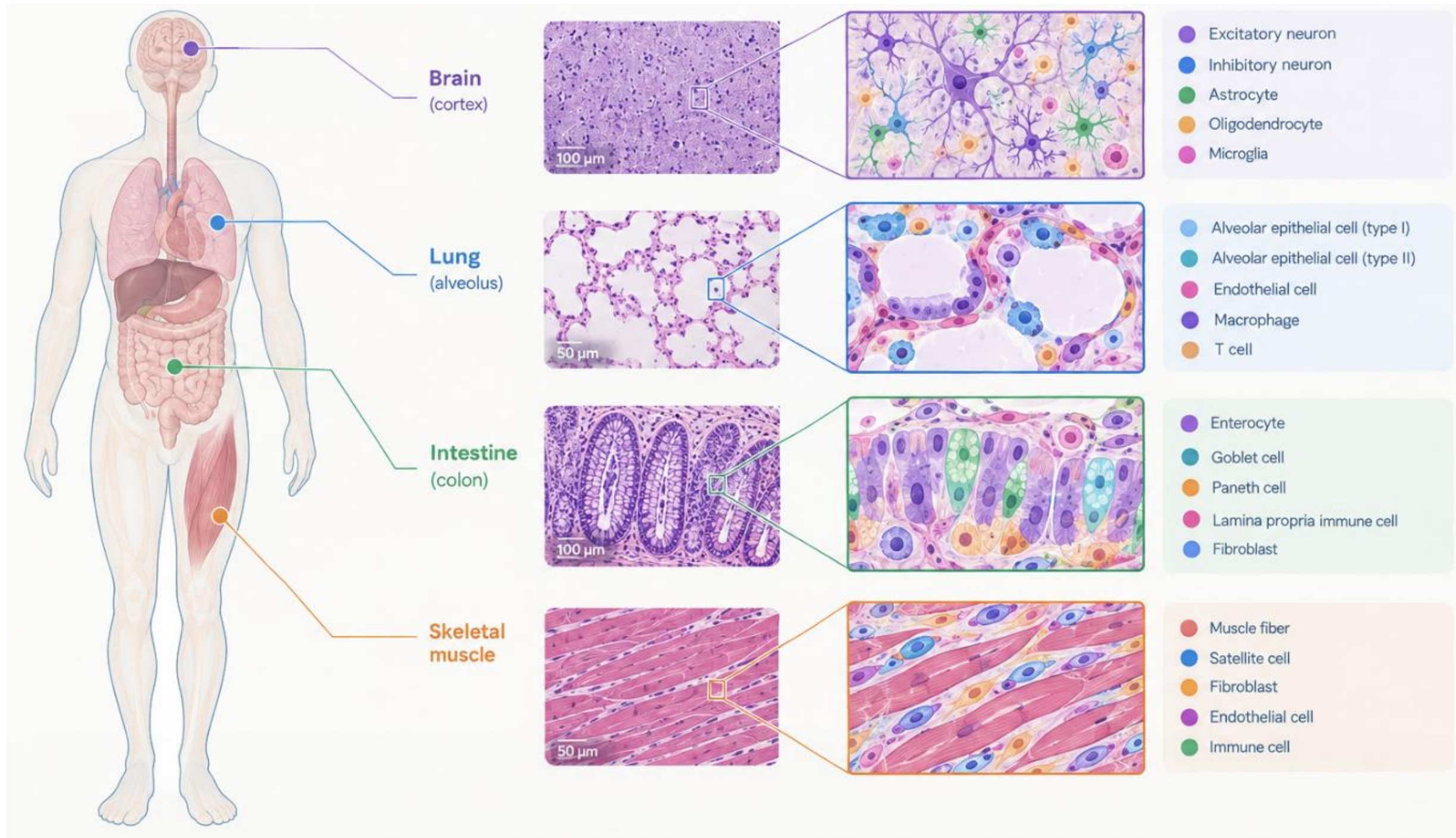
01

WHY SPACE?

What do we lose when we dissociate tissue?

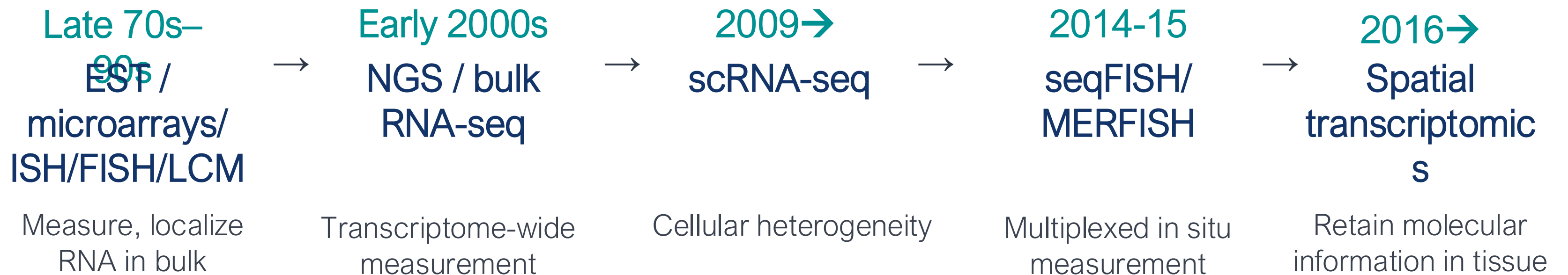


We know tissues are heterogeneous



Transcriptomics: Unraveling the nature of cellular identity

Evolution of transcriptomics + Historic milestones that led to spatial



The transition is from averaged tissue → individual cells → cells retained in tissue context.
Not simply “more, pretty data”. It is preserving a different kind of information

What is spatial transcriptomics? What problem are we trying to solve?

Three questions about the same tissue

BULK RNA-seq (\$) What is expressed?

averaged across tissue
(high sensitivity, low resolution)



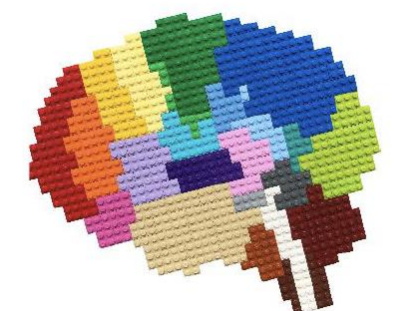
scRNA-seq (\$\$) Which cells express it?

cellular identity after dissociation
(high resolution + sensitivity, loss of spatial context)



SPATIAL (\$\$\$) Where is that expression?

Molecular state + location + tissue context
(resolution + sensitivity + spatial context)



What is spatial transcriptomics? What problem are we trying to solve?

Three questions about the same tissue

BULK RNA-seq (\$) What is expressed?

What molecular programs are present?



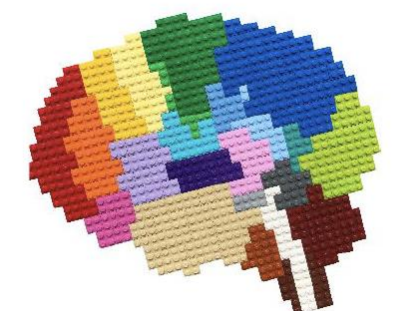
scRNA-seq (\$\$) Which cells express it?

Which cells/states carry them?

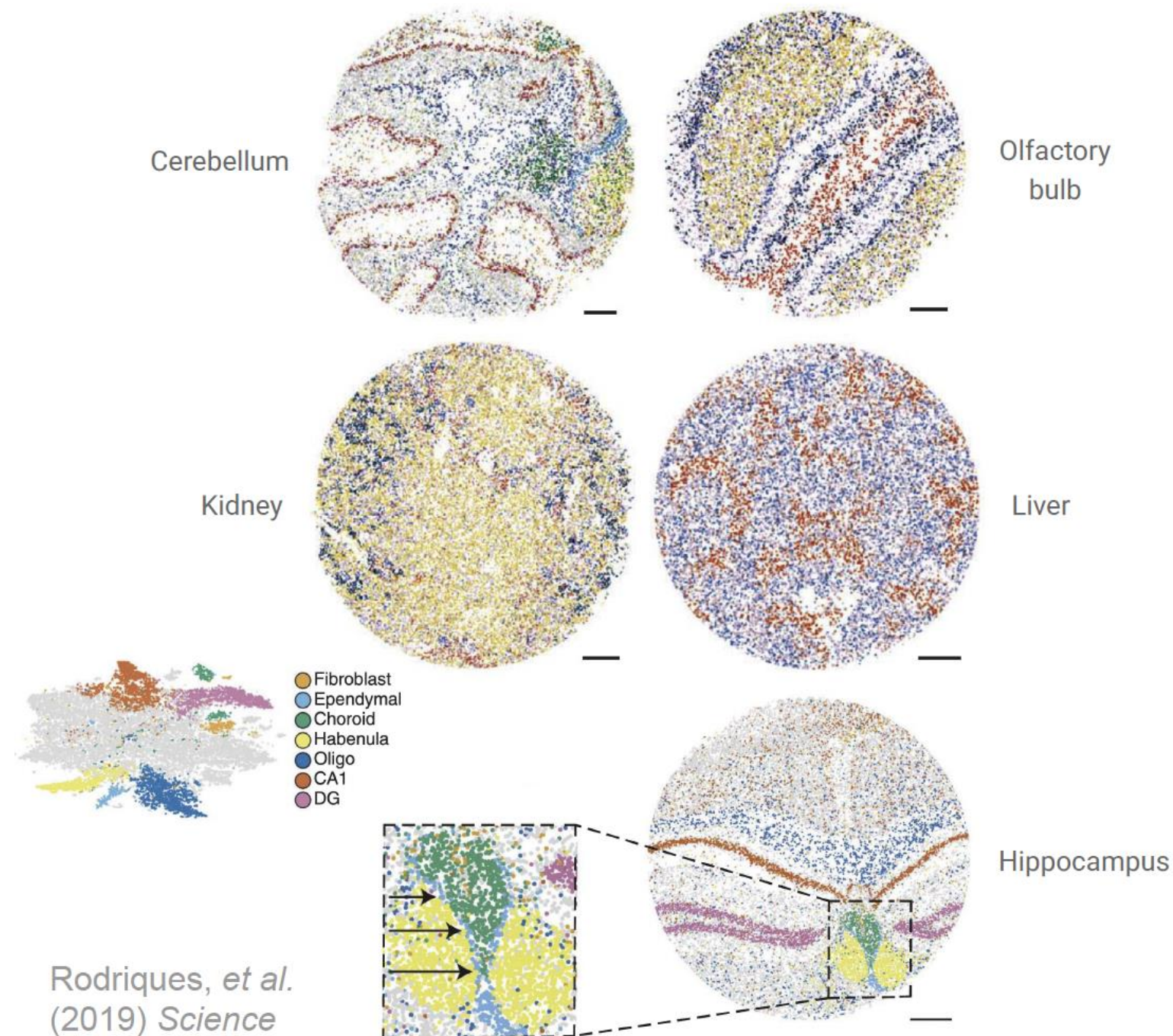


SPATIAL (\$\$\$) Where is that expression?

Where are those states organized within tissue?



Why study biology in the context of space?



Dissociation gives us cells – and removes relationships

To understand:

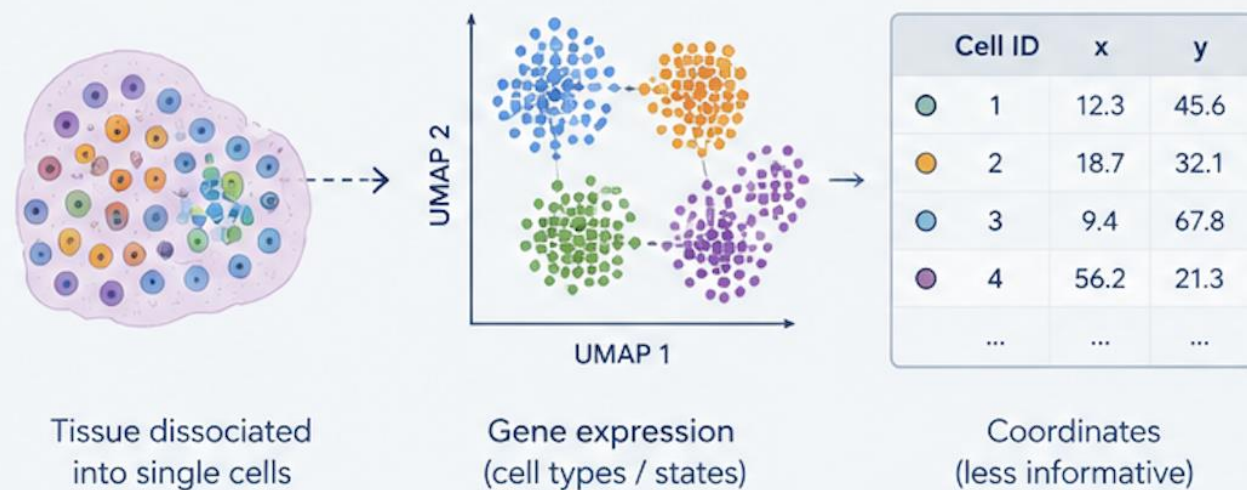
- cell-cell interactions in native environ
- map tissue architecture
- explore disease microenvironment
- identify spatial biomarkers
- visualize gradients and developmental cues

Spatial data generate hypotheses about relationships; validation is still required.

Spatial is not simply scRNAseq + co-ordinates

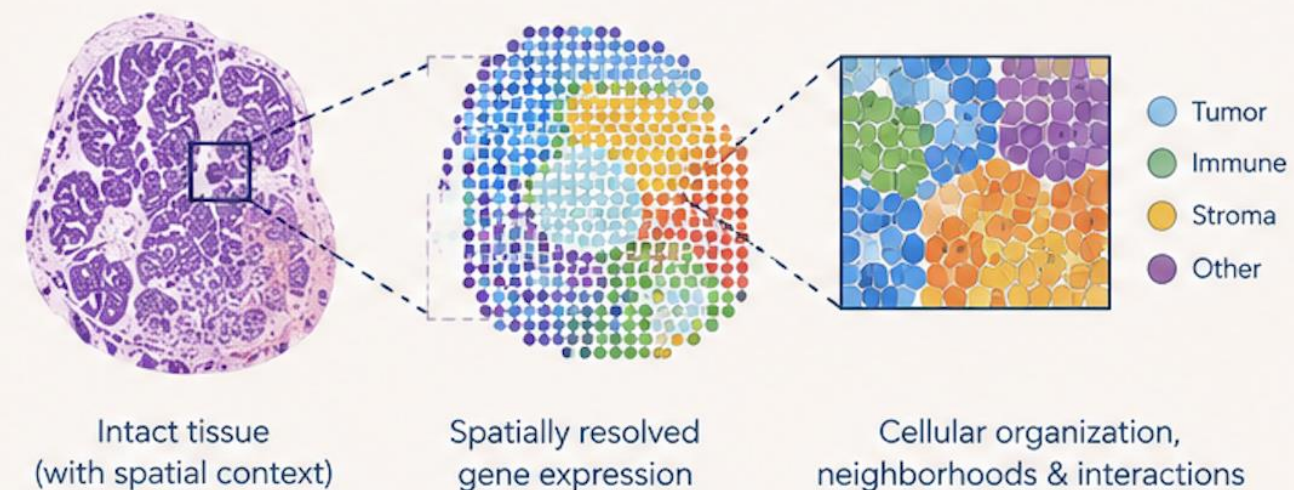
scRNA-seq + coordinates

Great for cell types and states. No spatial context.

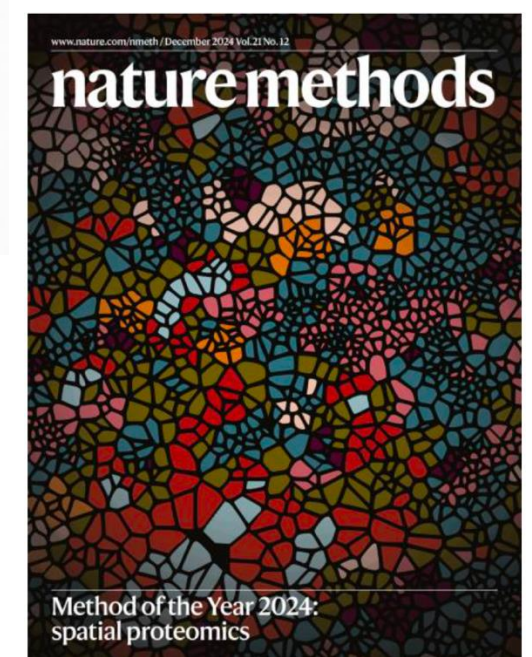


Spatial transcriptomics

Adds spatial context, preserves tissue architecture, and enables new biological insights.



Expression-space (scRNAseq) coordinates (e.g. UMAP, PCA) tell us **which cells are similar**.
Physical-space (spatial) coordinates tell us **where the measurement occurred**.



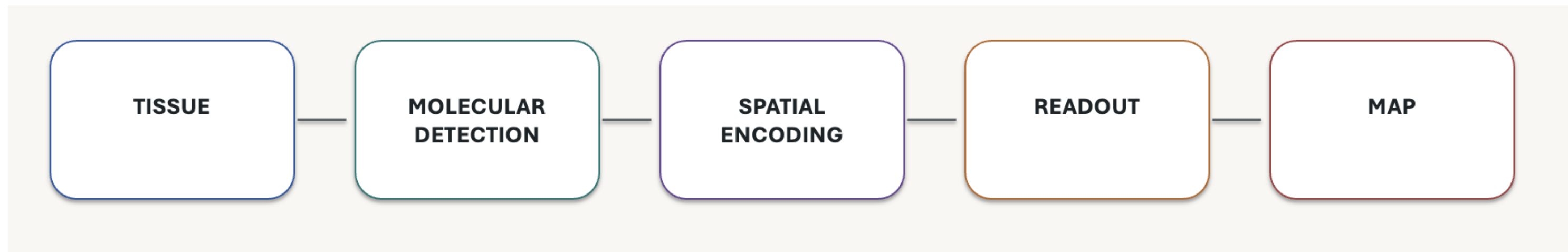
HOW DO WE MEASURE SPACE?

Two broad strategies preserve spatial information in different ways.



The spatial measurement problem

Connecting molecular identity to physical location



There are 2 dominant strategies & they differ in where the spatial information is encoded in the workflow

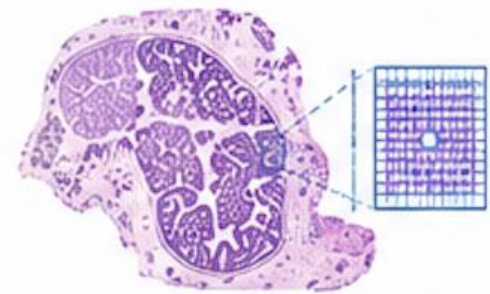
- Sequencing/capture-based
- Imaging/ in situ based

Both produce molecular identity + spatial coordinates + tissue context, but with different underlying assumptions

TIP: Remember where, physically & molecularly, the spatial information entered the experiment

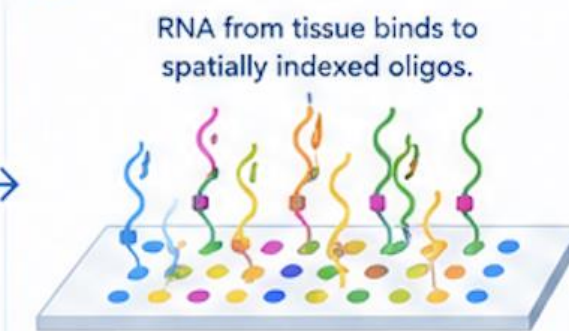
1 Sequencing-based route | Capture → Sequence → Reconstruct

1 Tissue section



A thin section of tissue is placed on a capture surface (e.g., Visium slide).

2 Capture & spatial barcoding



Spatially indexed oligonucleotides on the capture surface

3 Library prep & sequencing



We read the captured RNA, spatial barcode and UMIs in the sequencing data.

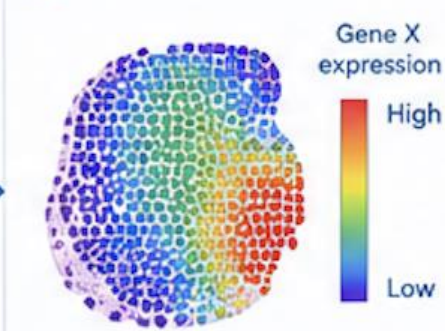
4 Computational reconstruction

Spatial unit × gene matrix

	Gene 1	Gene 2	Gene 3	...
Spot 1	12	5	0	...
Spot 2	3	8	1	...
Spot 3	0	2	14	...
...	...	...	...	...

Reads are assigned to spatial locations, generating a matrix of molecular counts.

5 Spatial expression map



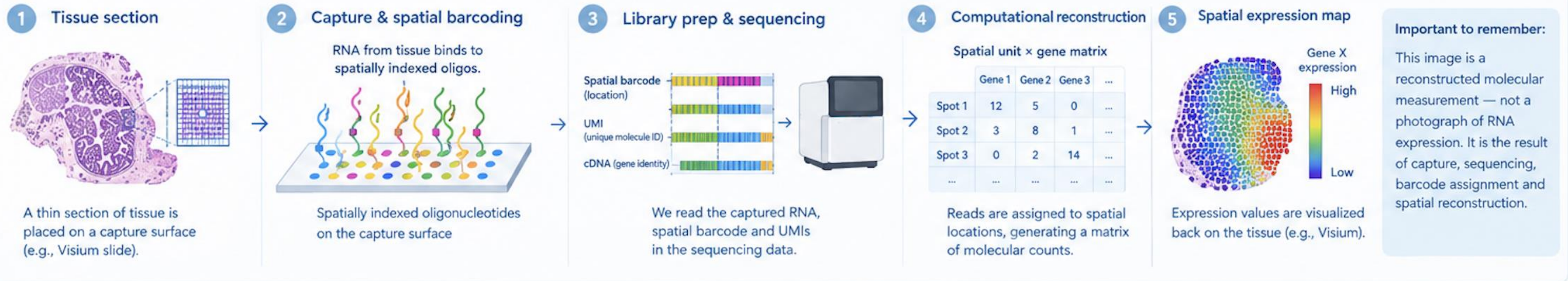
Expression values are visualized back on the tissue (e.g., Visium).

Important to remember:

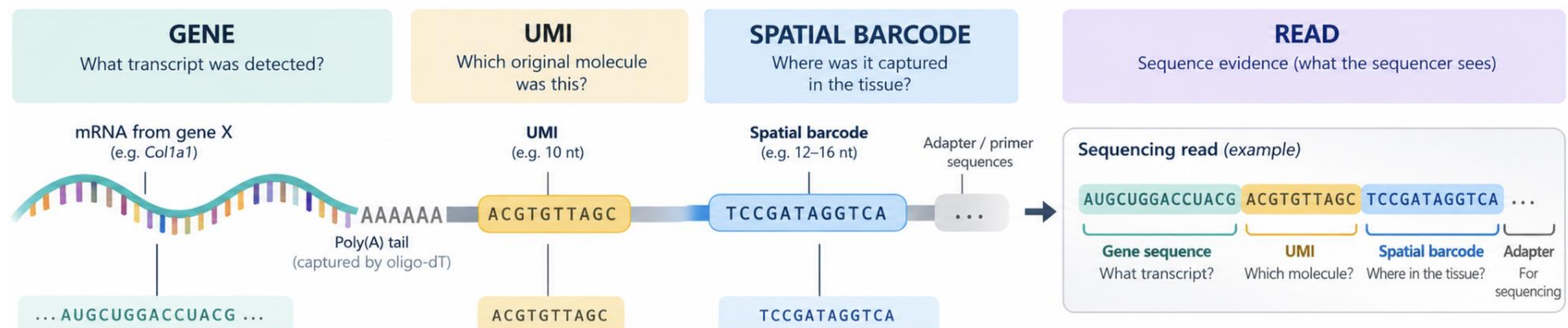
This image is a reconstructed molecular measurement — not a photograph of RNA expression. It is the result of capture, sequencing, barcode assignment and spatial reconstruction.

Key difference from ordinary RNAseq -- The biological tissue is converted into a spatially indexed molecular map i.e. molecular identity is linked to a physical location & carries the following information:

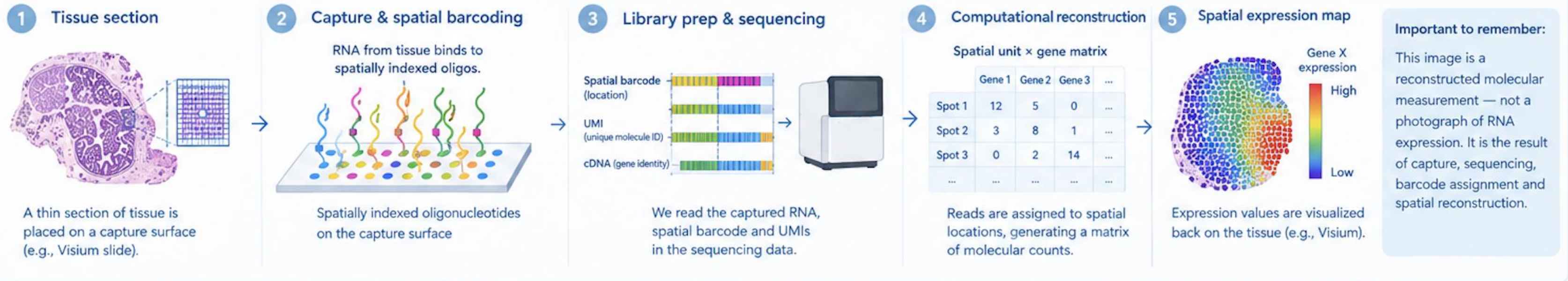
1 Sequencing-based route | Capture → Sequence → Reconstruct



Key difference from ordinary RNAseq -- The biological tissue is converted into a spatially indexed molecular index i.e. molecular identity is linked to a physical location & carries the following information:



1 Sequencing-based route | Capture → Sequence → Reconstruct

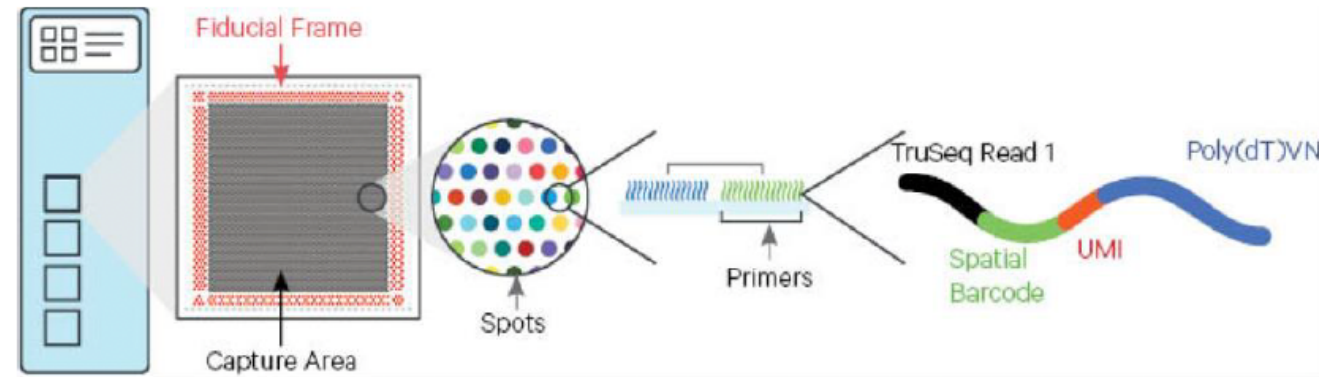


Broadly speaking: RNA captured via Poly(A) tail & sequenced, but can be probe based

2 main sub-categories:

- **Spatial sequencing** (10x Visium HD, Curio Seeker, BGI STOmics—Stereo-seq etc)
- **Spatial tagging** (Dbit-seq, AtlasXomics, Curio Trekker etc)
- Total mRNA capture and often “whole transcriptome” i.e. “discovery/ unbiased/ untargeted”
- But can be targeted
- Species agnostic (mRNA from any species with Poly(A) can be captured) – great for non-human, non-mouse samples

10x Visium



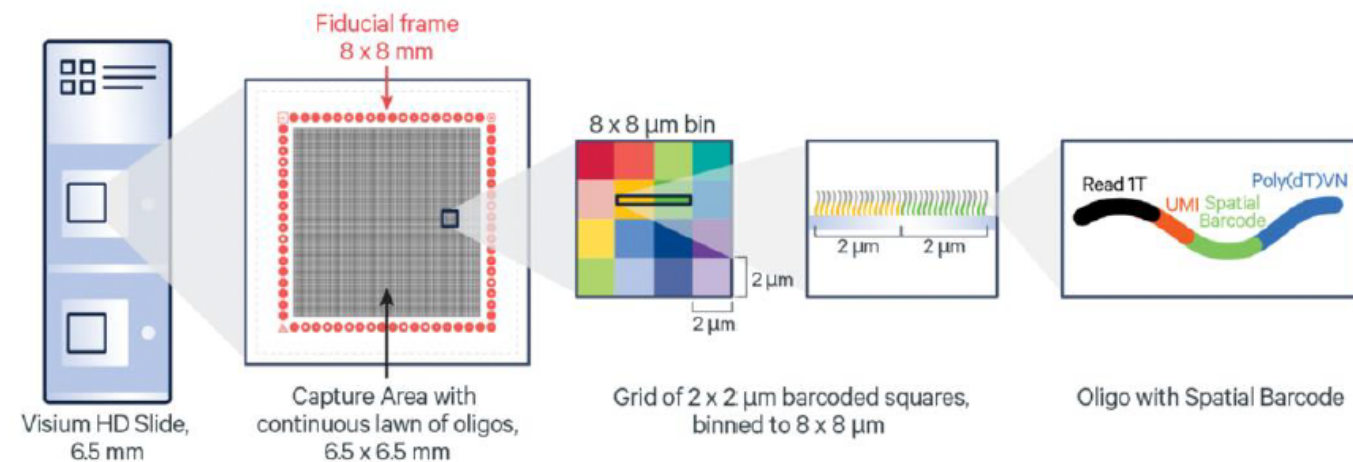
• Visium V1

- 3' poly(A) captured based
- Fresh frozen samples
- Diverse species
- 55µm barcoded spot (1-10 cells)

• Visium V2

- Probe based chemistry
- Fresh frozen, fixed frozen, FFPE
- Human and Mouse
- 55µm barcoded spot (1-10 cells)

10x Visium HD



• Visium HD

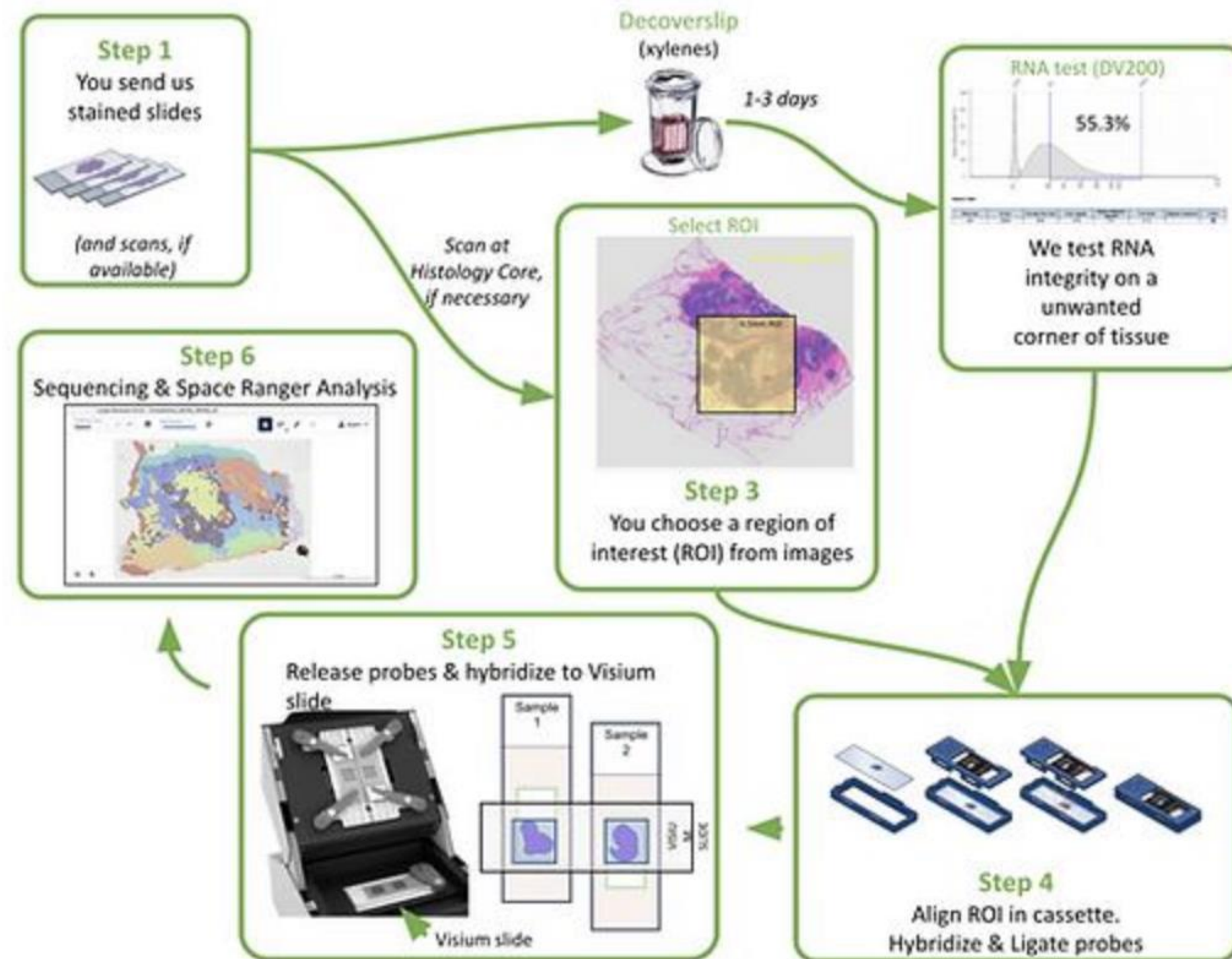
- Probe based chemistry
- Fresh frozen, fixed frozen, FFPE
- Human and Mouse
- 2µm barcoded spot

• Visium HD 3'

- 3' poly(A) captured based
- Fresh frozen samples
- Diverse species
- 2µm barcoded spot

How does Visium turn tissue into a spatial dataset?

Tissue → Capture → Library → Sequence → Reconstruct → Map



Case study: mapping the colorectal tumor microenvironment

de Oliveira et al., 2025 – the dataset behind this workshop

nature genetics



Article

<https://doi.org/10.1038/s41588-025-02193-3>

High-definition spatial transcriptomic profiling of immune cell populations in colorectal cancer

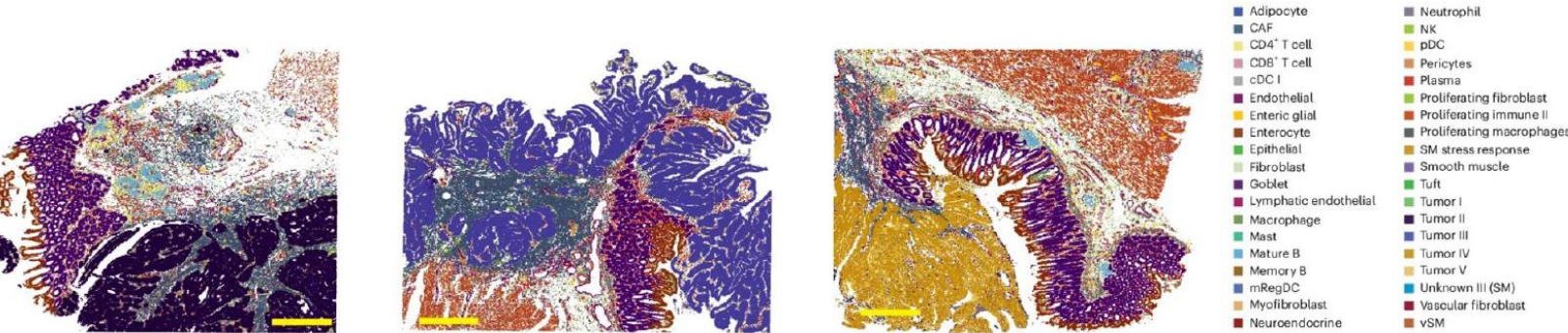
Received: 13 June 2024

Accepted: 14 April 2025

Published online: 5 June 2025

Check for updates

Michelli Faria de Oliveira^{1,2}, Juan Pablo Romero^{1,2}, Meili Chung^{1,2}, Stephen R. Williams¹, Andrew D. Gottscho¹, Anushka Gupta¹, Susan E. Pilipauskas¹, Seayar Mohabbat¹, Nandhini Raman¹, David J. Sukovich¹, David M. Patterson¹, Visium HD Development Team* & Sarah E. B. Taylor¹✉



THE QUESTION

Where are distinct immune and tumor states located within the CRC tissue architecture?

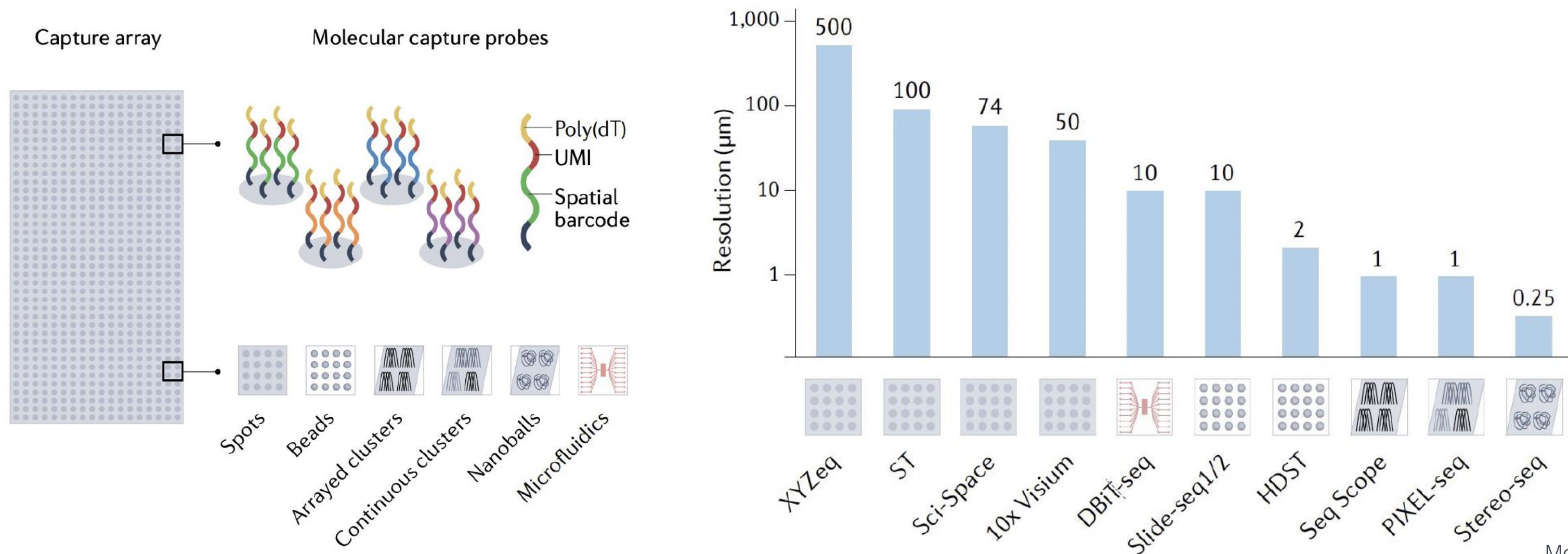
DISCOVER → MAP → INTERPRET

- Single-cell data provide a reference repertoire.
- Visium HD preserves tissue context at high-density spatial sampling.
- Spatial analysis asks how those states are organized.

Sequencing-based spatial methods use different physical architectures

Spots | Beads | Arrays | Nanoballs | Microfluidics

Different implementations change sampling, coverage, capture and reconstruction — with trade-offs.
Look past the label and ask: what is actually being measured?

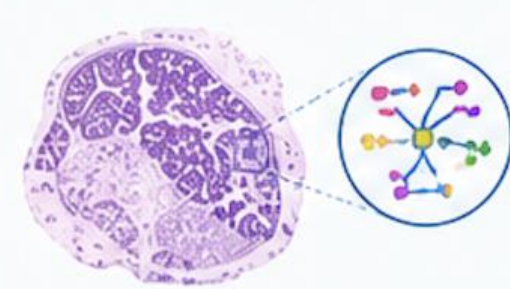


2

Imaging-based route

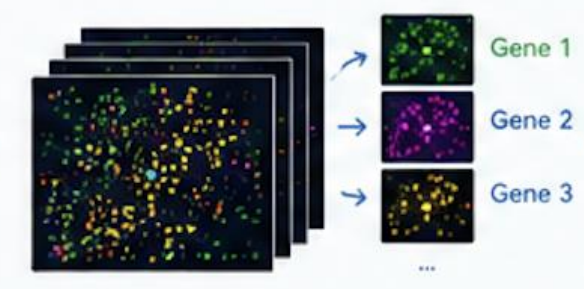
Detect → Image → Decode → Assign

1 Probe / detection chemistry



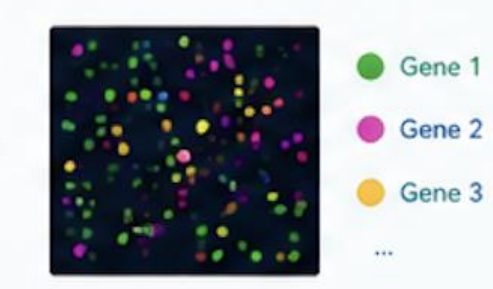
Fluorescent probes or in situ chemistry bind to target transcripts in the tissue.

2 Image acquisition



We acquire images with spatially resolved molecular signals.

3 Decode signals



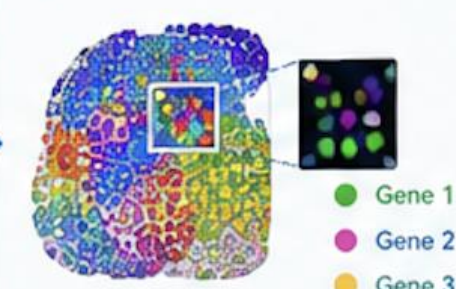
Signals are decoded so we know which transcript each signal represents.

4 Segment & assign to cells/regions



Segmentation and transcript assignment link signals to cells or tissue regions.

5 Molecular + morphology



The molecular signal and tissue morphology are measured together.

6 Cell-by-gene matrix

	Cell 1	Cell 2	Cell 3	...
Gene 1	12	3	7	...
Gene 2	5	14	2	...
Gene 3	0	6	11	...
...	...	...	...	...

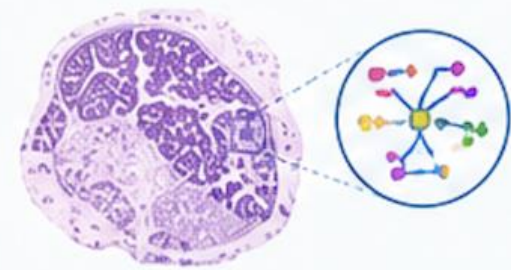
Segmentation and assignment are needed to generate a cell-by-gene matrix.

2

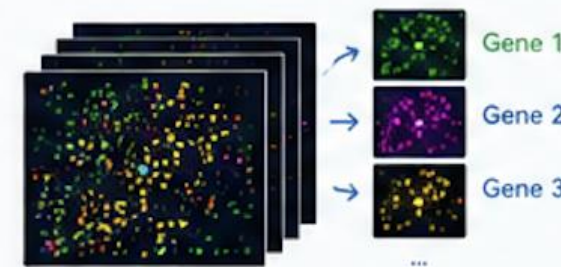
Imaging-based route

Detect → Image → Decode → Assign

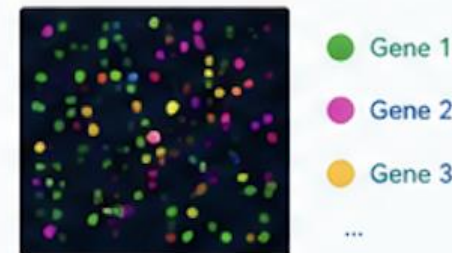
1 Probe / detection chemistry



2 Image acquisition



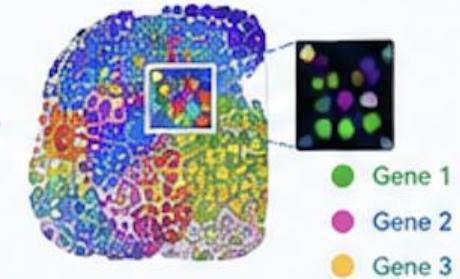
3 Decode signals



4 Segment & assign to cells/regions



5 Molecular + morphology



6 Cell-by-gene matrix

	Cell 1	Cell 2	Cell 3	...
Gene 1	12	3	7	...
Gene 2	5	14	2	...
Gene 3	0	6	11	...
...	...	...	...	...

Segmentation and assignment are needed to generate a cell-by-gene matrix.

RNA detected through repeated cycles of fluorescence imaging

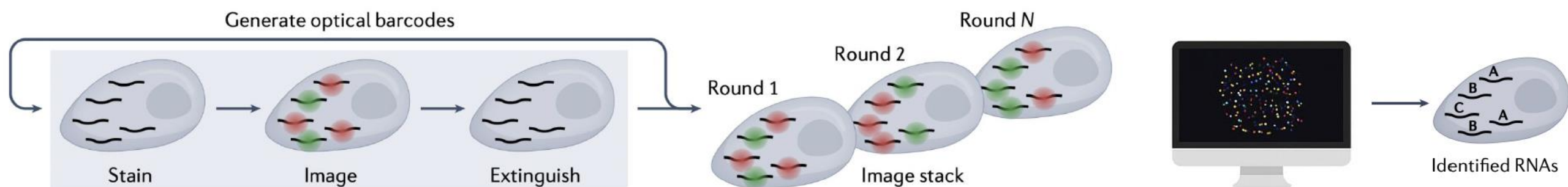
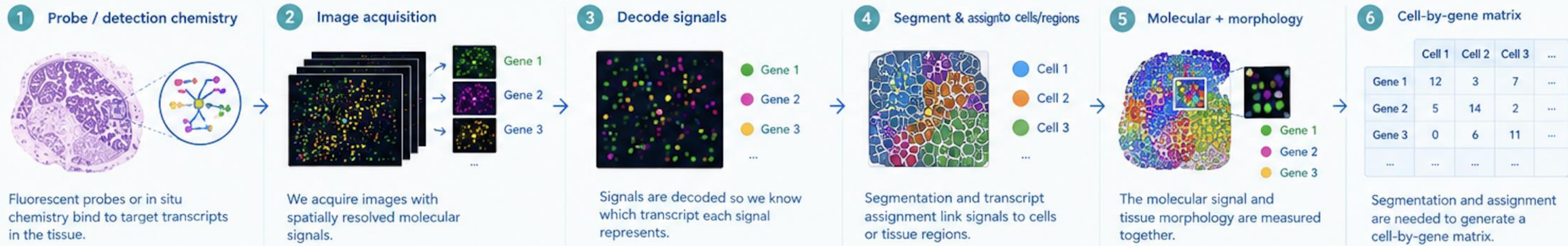
2 main sub-categories:

- **Barcoded FISH** (Vizgen MERSCOPE, 10x Xenium, Nanostring/Bruker CosMx etc)
- **In situ sequencing** (Singular Genomics G4X, StellarOmics etc)
- “Targeted” transcriptome capture: probes against genes of interest, therefore, usually not “whole-genome”
- Probe panel has to be designed beforehand
- Probes allow very fine resolution, often at single-molecule level
- Not species agnostic (limited to e.g. mice or human)

2

Imaging-based route

Detect → Image → Decode → Assign



Generate Optical Barcodes

Probes bind to targets →
signal read out via single- or
multi-color fluorescence

Sequential Imaging + Signal Removal

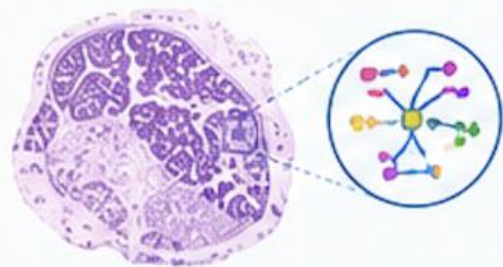
Multiple rounds of
fluorescence imaging w/
different barcodes

Image Processing

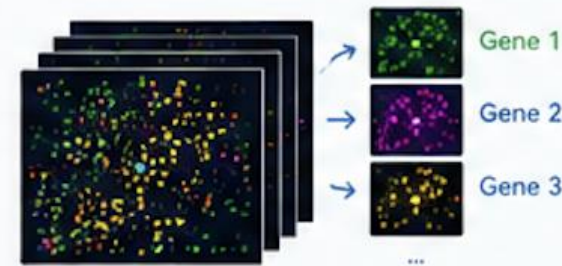
Computational decoding
identifies RNA molecules

2 Imaging-based route | Detect → Image → Decode → Assign

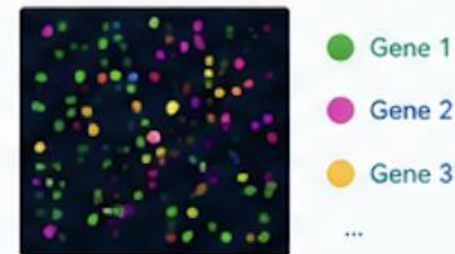
1 Probe / detection chemistry



2 Image acquisition



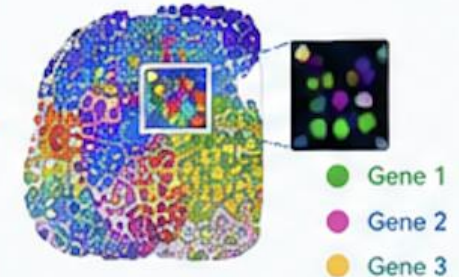
3 Decode signals



4 Segment & assign to cells/regions



5 Molecular + morphology



6 Cell-by-gene matrix

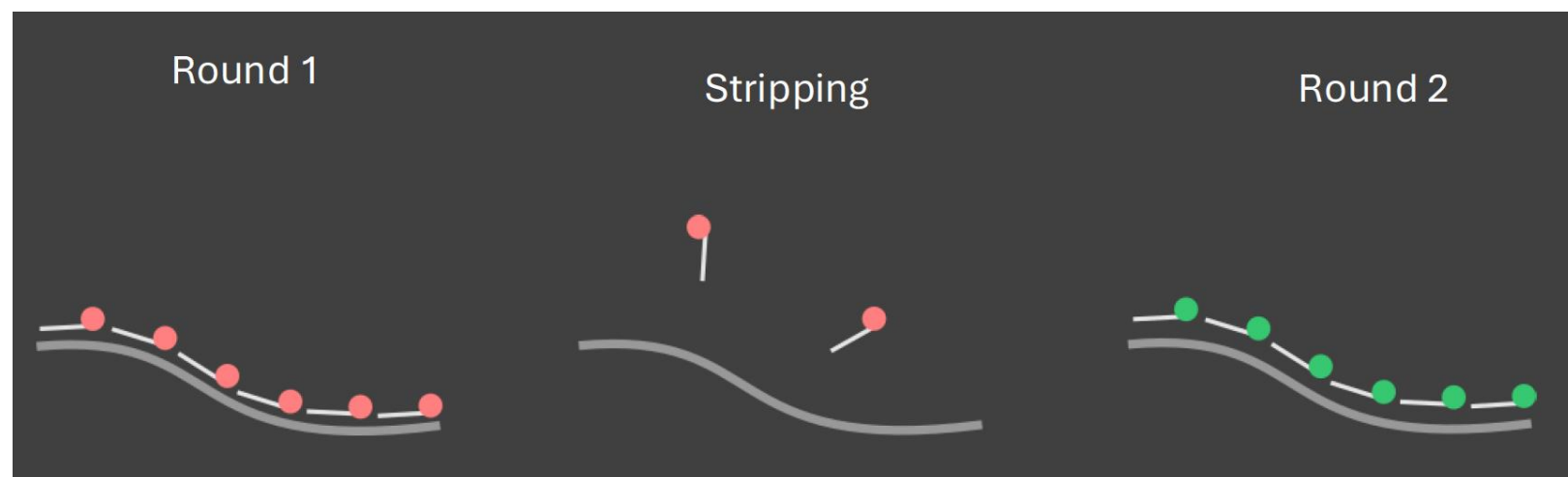
	Cell 1	Cell 2	Cell 3	...
Gene 1	12	3	7	...
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Gene 3	0	6	11	...
...	...	...	...	...

Segmentation and assignment are needed to generate a cell-by-gene matrix.

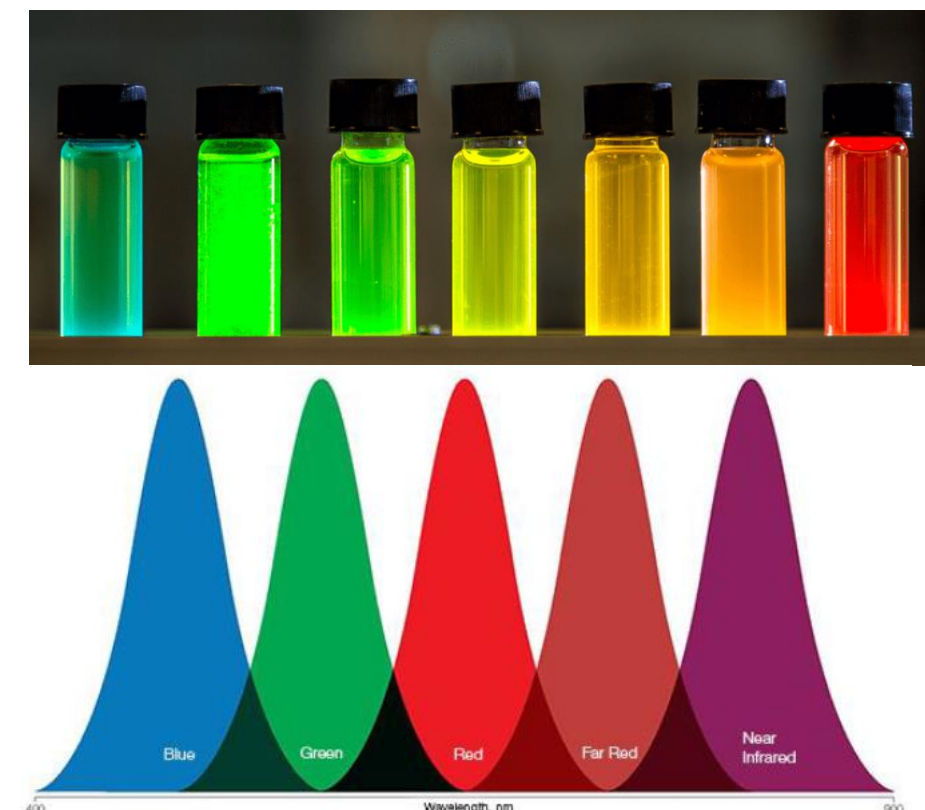
Why is repeated staining of optical barcodes necessary?

For technical reasons and for scaling through put

e.g. Limited fluorophores (4-7), but ~20,000 genes



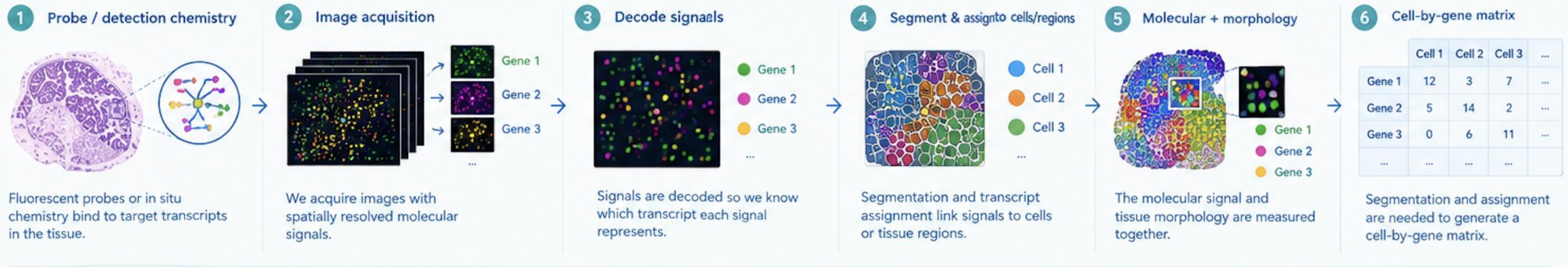
P.C. Lars Borm



2

Imaging-based route

Detect → Image → Decode → Assign



Scaling = (No. of fluorophores)^{number of rounds} = large barcode space

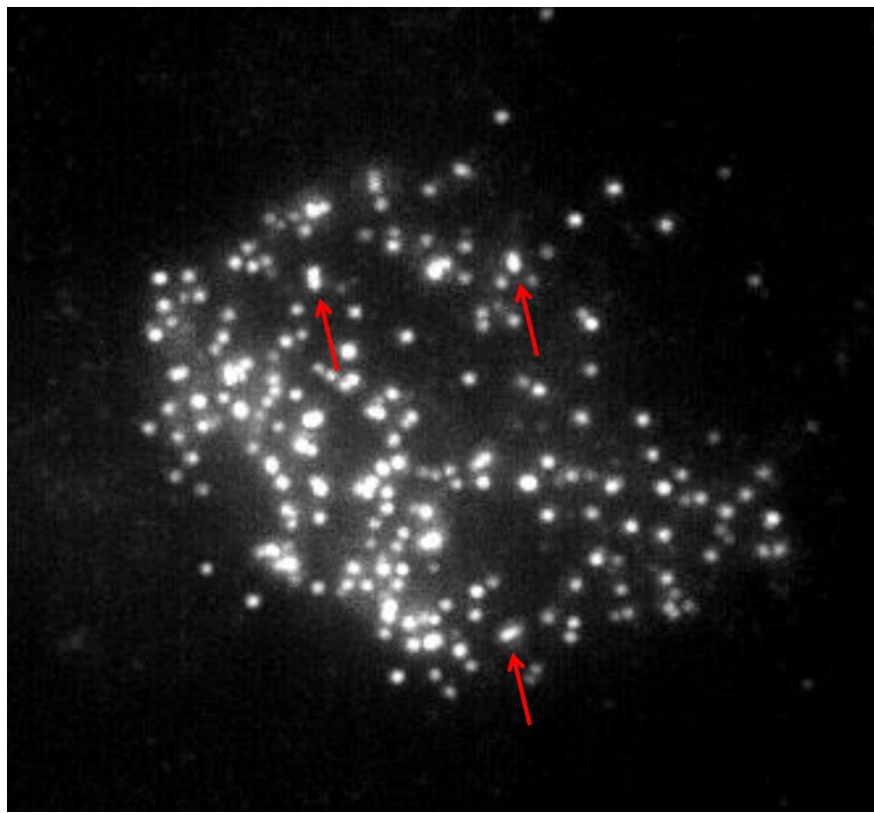
E.g. 4⁸ = 65,536 (i.e. 3x human transcriptome)

A small number of fluorophores can encode many transcript identities through repeated imaging rounds

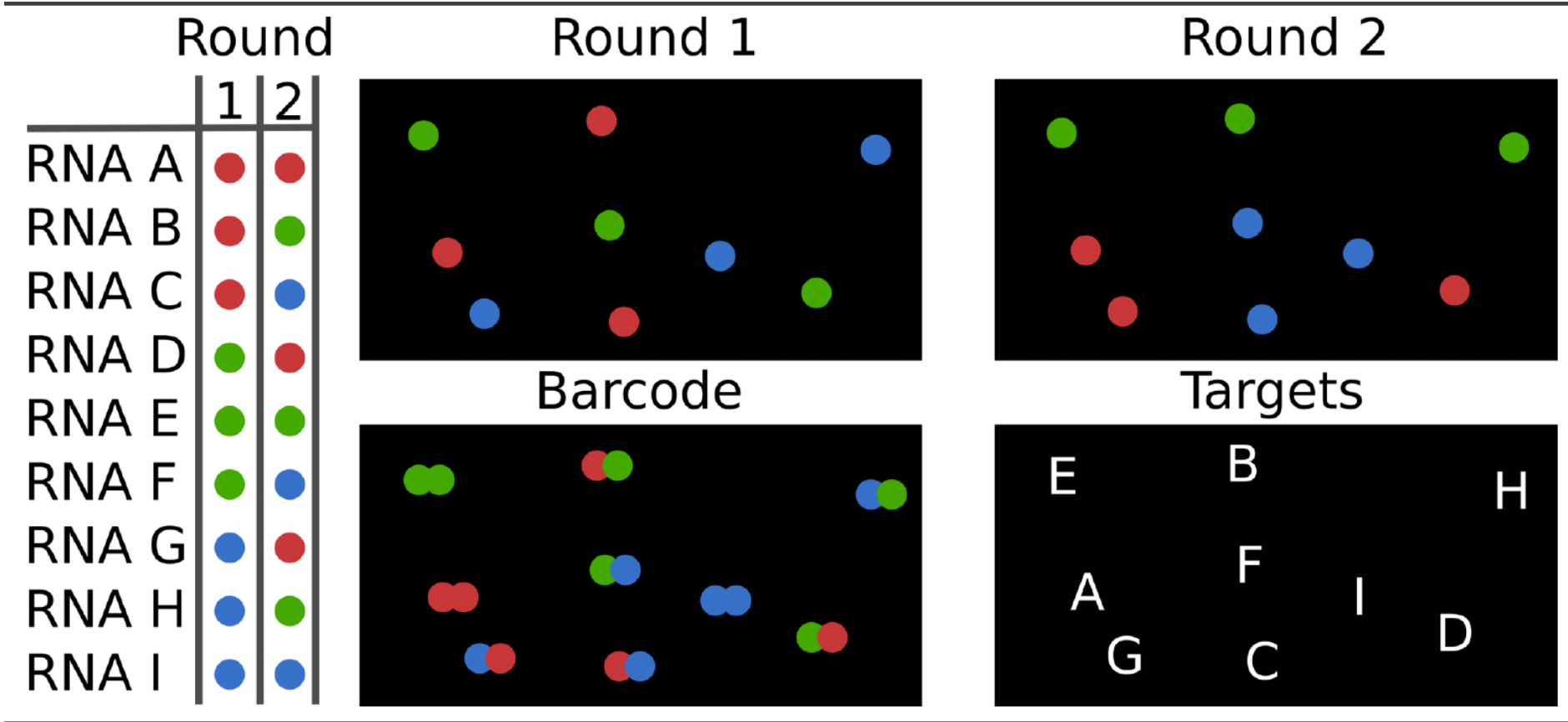
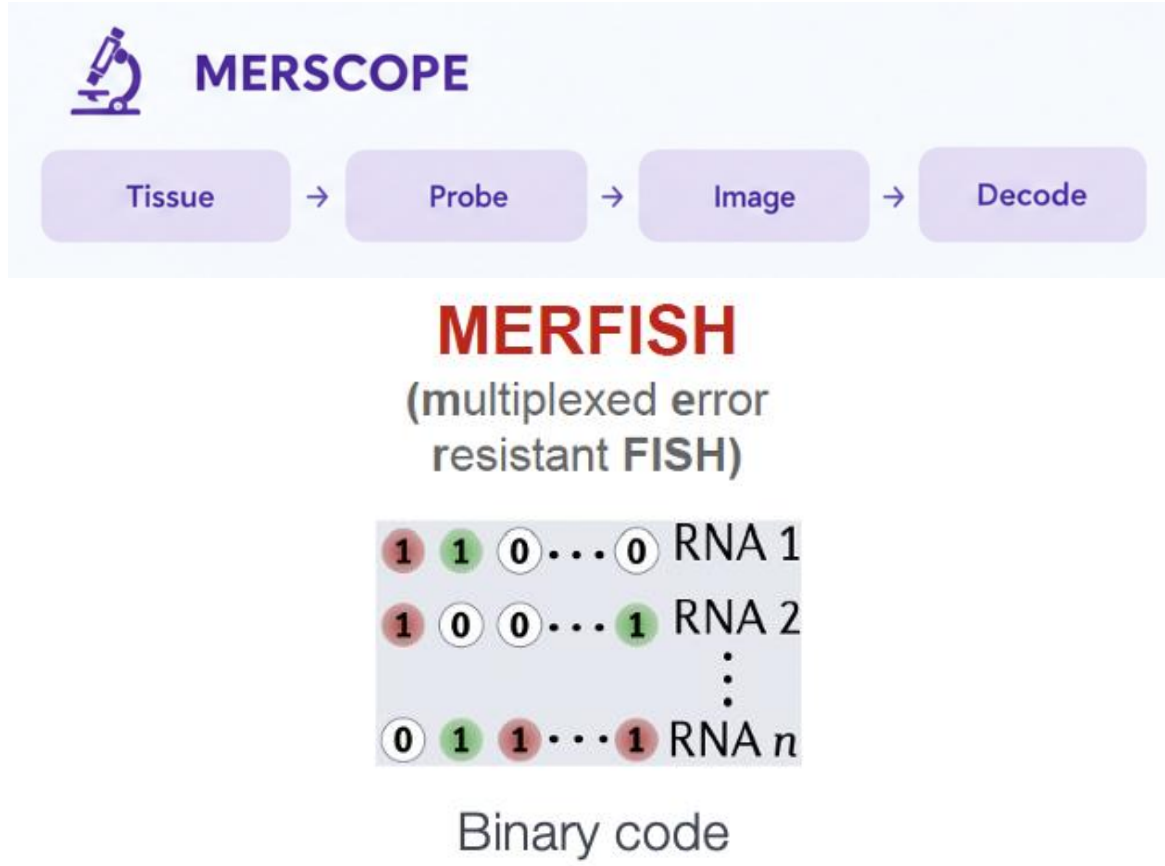
Problems:

Optical crowding (hit microscope resolution limits) & error rates

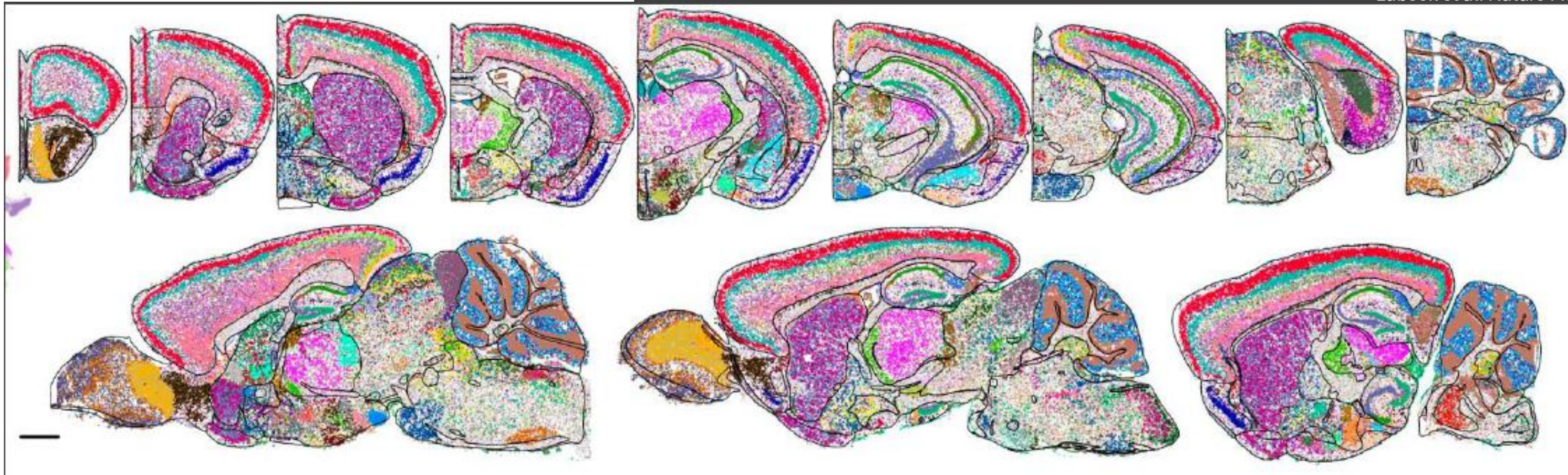
Solution: Sparce, error robust barcodes (introduced by MERFISH)



P.C. Lars Borm



Lubeck et al. Nature Methods 2014

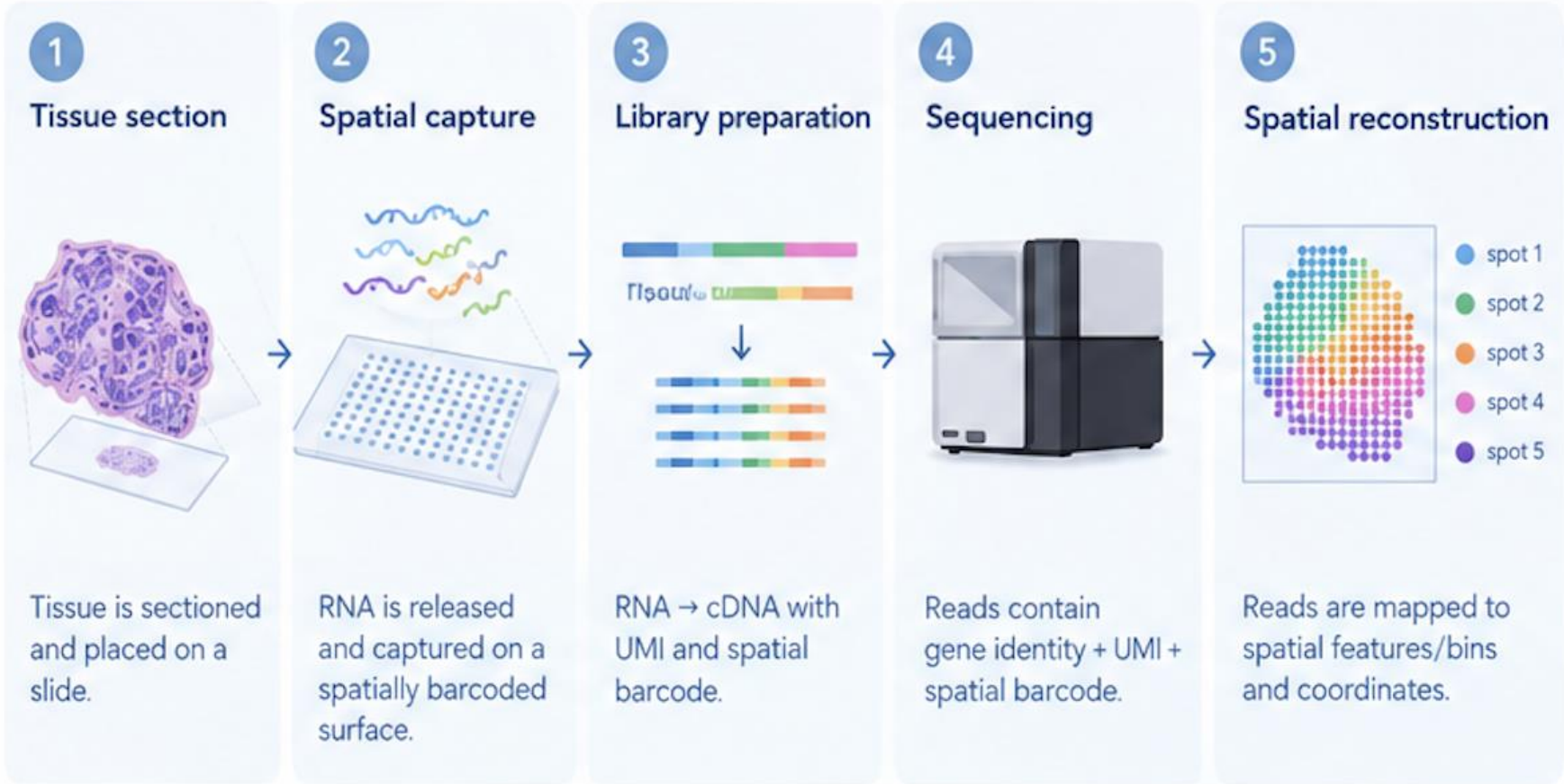


Zhang et al., 2023, Whole atlas of the mouse brain

Visium HD

(Sequencing-based)

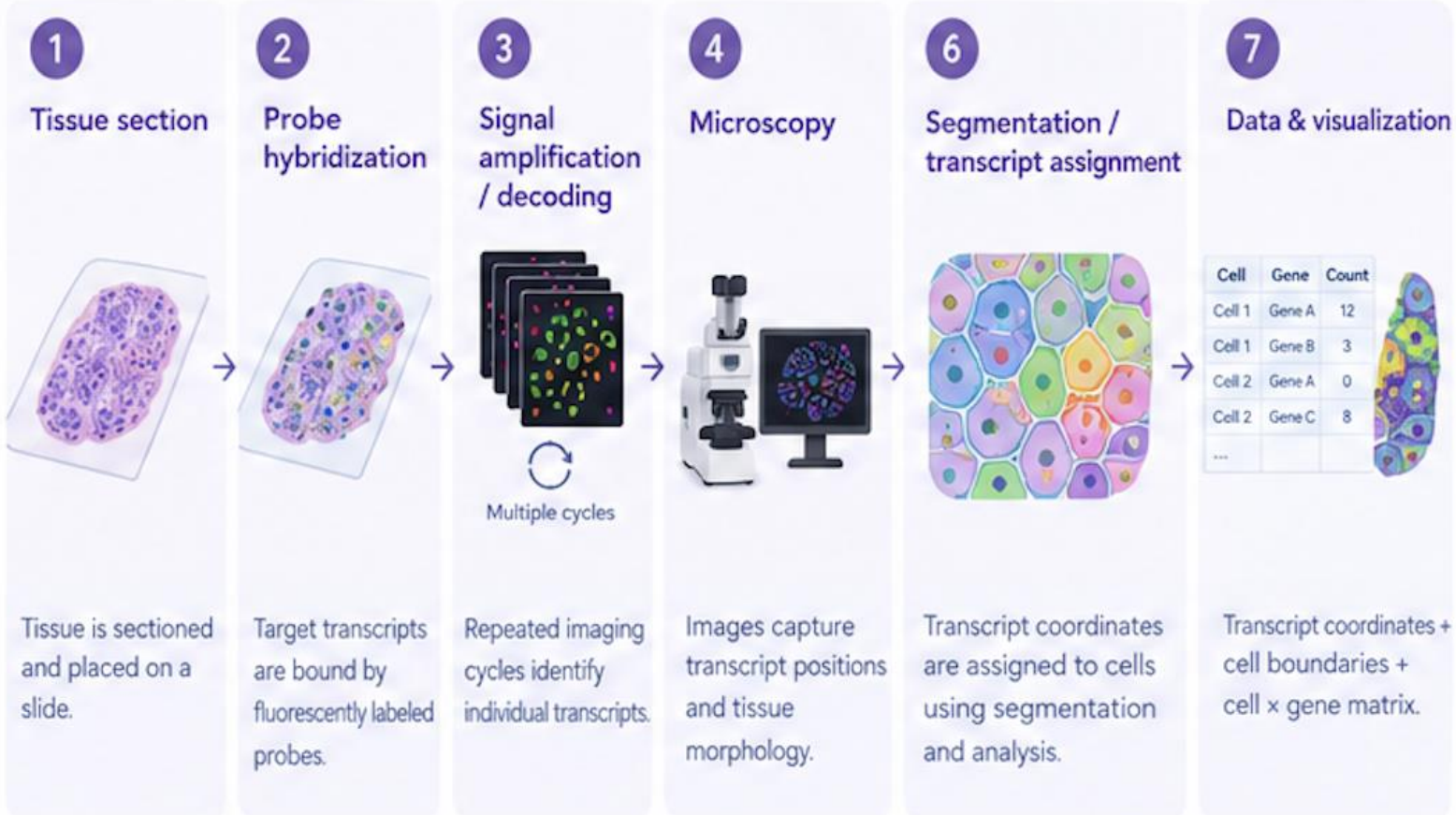
Whole-transcriptome | High-density spatial sampling



MERSCOPE

(Imaging-based)

Targeted / high-plex panel | Single-molecule / subcellular



Best for: broad discovery, tissue architecture, large area coverage



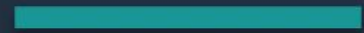
MERSCOPE



Best for: cellular interactions, fine spatial detail, molecular localization

WHAT DOES “RESOLUTION” MEAN?

Stop treating resolution as one magic number. Ask what spatial unit the measurement can distinguish.

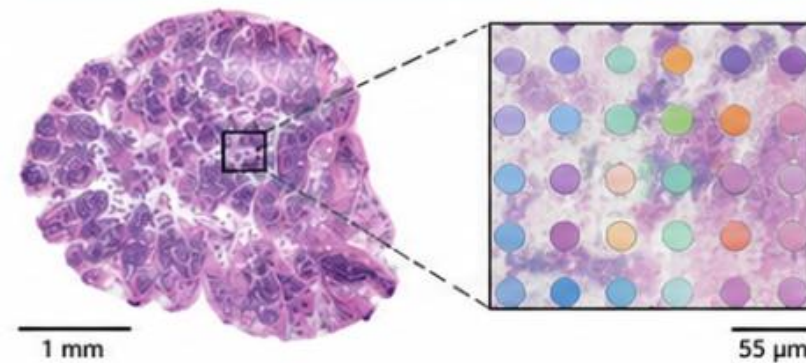


What is the spatial unit?

The measurement unit determines what you can ask.

SPOT

(e.g. Visium)

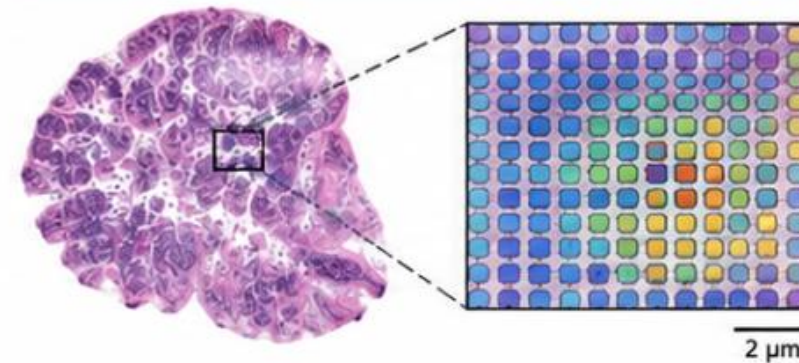


What is measured?

A predefined capture area
(e.g. 55 µm spot).

BIN

(e.g. Visium HD)

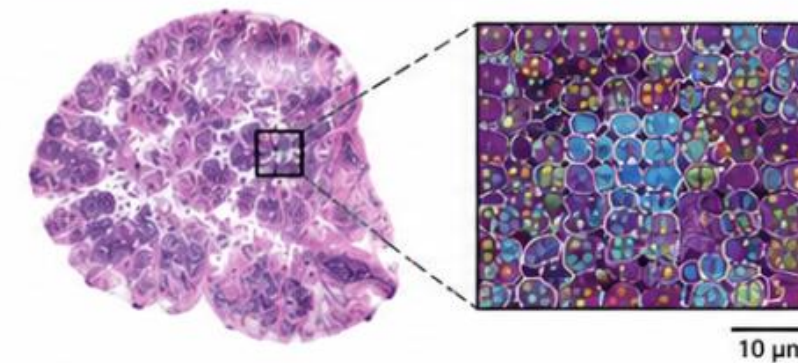


What is measured?

A smaller spatial feature
(analytical bin, e.g. 2 µm).

SPATIAL BARCODE

(e.g. MERFISH, Xenium)

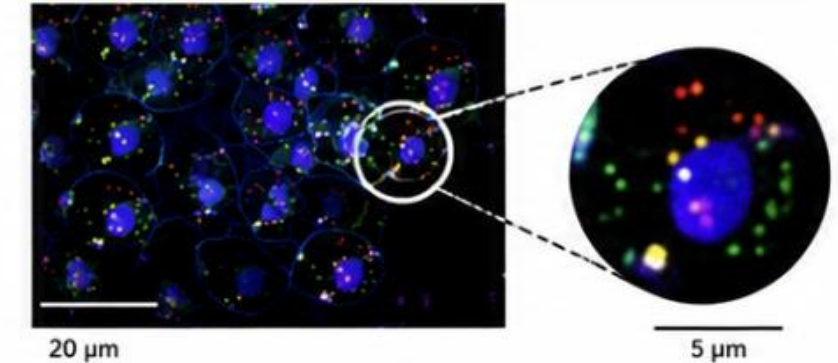


What is measured?

Individual RNA molecules
(with detected position).

MOLECULE / CELL

(e.g. imaging-based)



What is measured?

RNA signal detected *in situ*
(at cellular/subcellular resolution).

← Same tissue. Different spatial units. Different questions. →

What exactly does one measurement represent?

Every spatial technology defines a spatial measurement unit. That unit may be a spot, bin, molecule, or inferred cell. And only some of those correspond directly to biological cells.

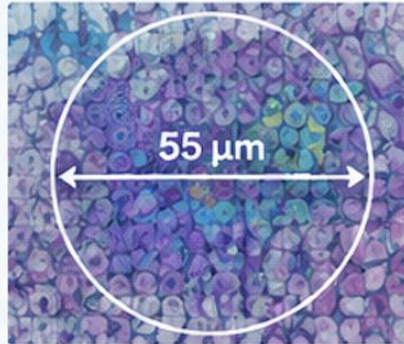
A spatial measurement unit is not the same thing as a biological unit.

An $X \mu\text{m}$ feature $\neq$ an $X \mu\text{m}$ cell

Spatial resolution describes the measurement — not the biology.

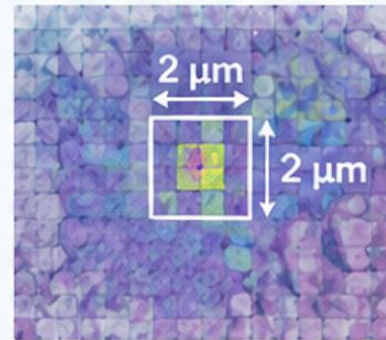
MEASUREMENT UNITS

Spot (e.g. $55 \mu\text{m}$)



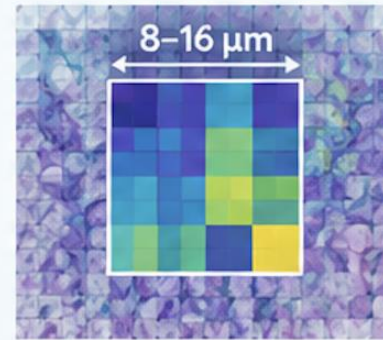
A predefined capture area that can contain multiple cells.

Feature (e.g. $2 \times 2 \mu\text{m}$)



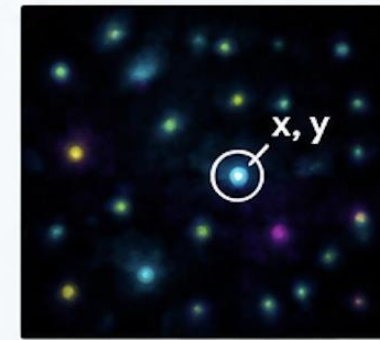
A spatial feature that may sample part of a cell, multiple cells, or a boundary.

Bin (e.g. 8 or $16 \mu\text{m}$)



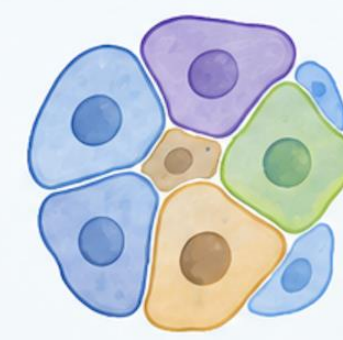
A computational grouping of smaller features.

Transcript coordinate



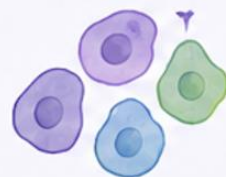
The location of a detected molecule (transcript).

Cell (biological unit)



A biological unit with a boundary that may need to be measured or inferred.

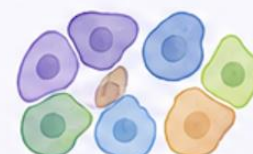
BIOLOGICAL UNITS



Multiple cells



Part of a cell
or cell boundary



Multiple cells



Molecules



One cell
(or a cell region)

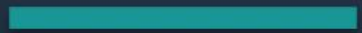


Measurement size $\neq$ biological unit.

A $2 \mu\text{m}$ feature can be smaller than, comparable to, or intersecting parts of cells. It does **not** magically become a cellular measurement. Likewise, a $55 \mu\text{m}$ spot is obviously not a $55 \mu\text{m}$ cell.

HOW DO YOU CHOOSE A TECHNOLOGY?

Start with the biological question — then list the information you must preserve.



Choosing the right technology for your question

Question 1: Discovery (e.g. new biomarker) or Targeted (e.g. validating a gene panel)?

Question 2: What resolution do you actually need?

Subcellular, cellular or tissue-level, scRNAseq reference dataset available

Question 3: What tissue do you have?

Fresh/frozen, FFPE, archival tissue, tissue size, RNA integrity, sectioning

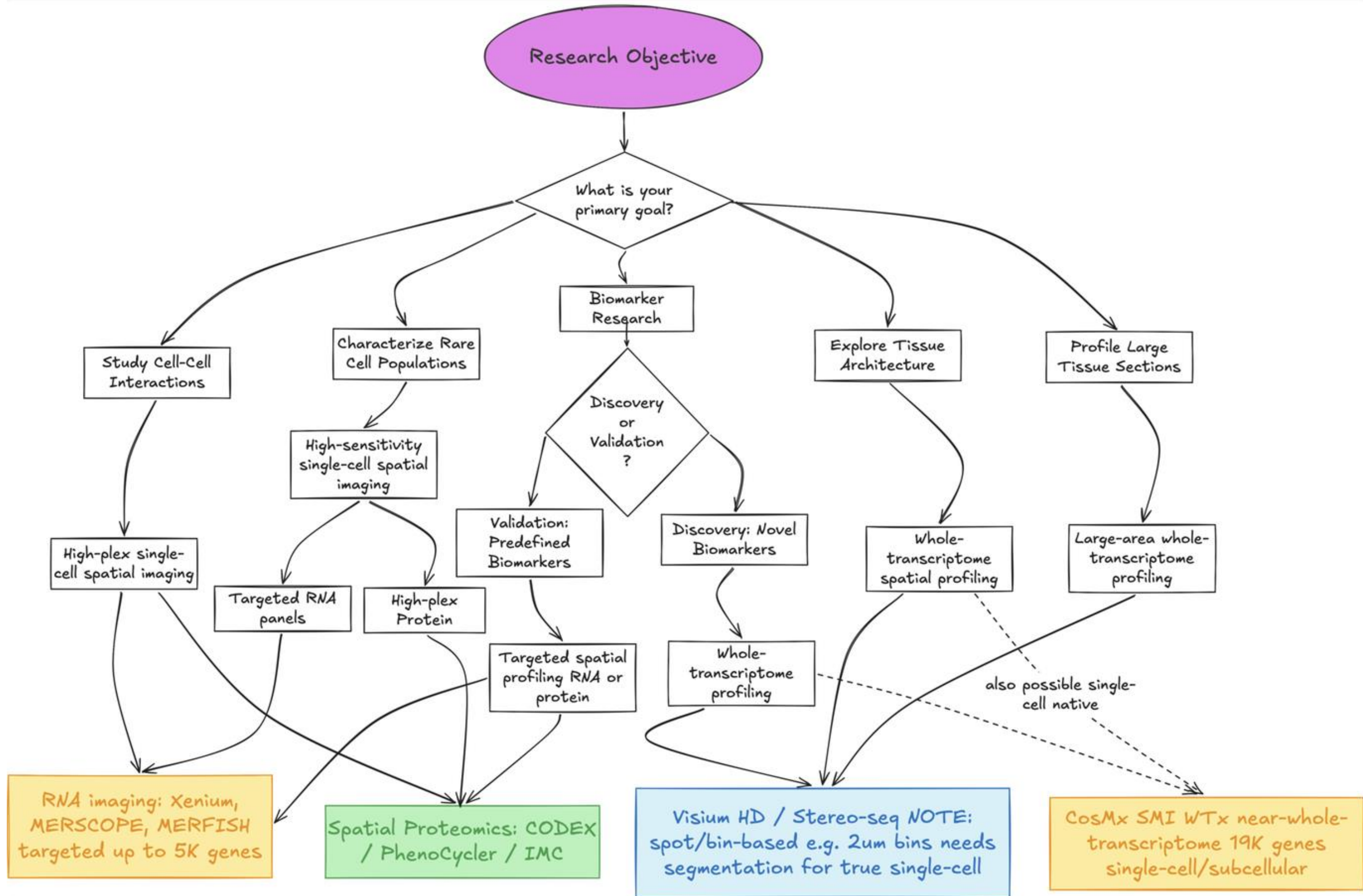
Question 4: How many samples?

Pilot (method dev), Mechanistic (Deep tissue biology), Cohort (replication + scale), Atlas (large scale mapping)

Question 5: What else must you measure? RNA, Protein, Epigenome, Genetics, Morphology

Question 6: do you have access to other required infrastructure and costs?

Computational, histology, pathology, microscopy, spatial platforms

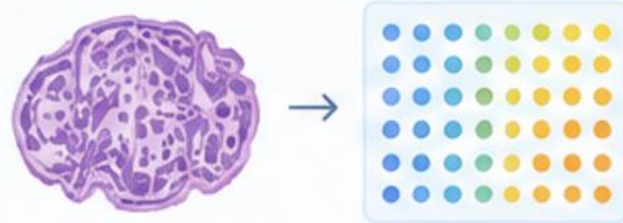


No spatial technology does everything

Different technologies, different strengths — and trade-offs.

Visium

Sequencing-based (capture)



 **TRANSCRIPTOME**
Whole-transcriptome

 **SPATIAL SCALE**
Spot-based ($\approx 50\text{--}100\ \mu\text{m}$)

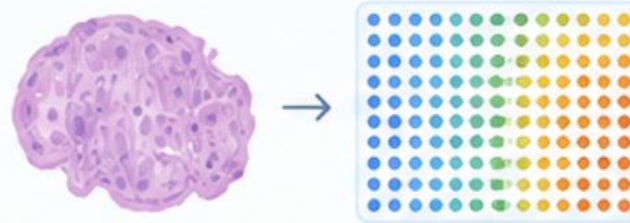
 **TISSUE CONTEXT**
Large tissue areas

 **BEST WHEN...**
Broad discovery & tissue architecture

 **KEY TRADE-OFF**
Coarser spatial resolution; individual cells may span multiple spots (requires deconvolution).

Visium HD

Sequencing-based (capture)



 **TRANSCRIPTOME**
Whole-transcriptome

 **SPATIAL SCALE**
 $2\ \mu\text{m}$ features (analysis often uses larger bins)

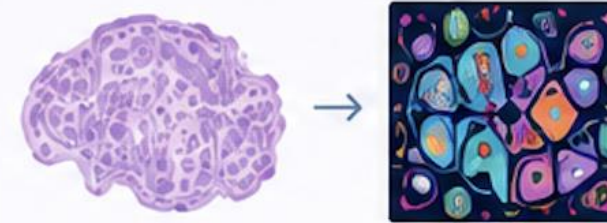
 **TISSUE CONTEXT**
Large tissue areas

 **BEST WHEN...**
Broad discovery + finer spatial sampling

 **KEY TRADE-OFF**
More data and analysis complexity; resolution depends on binning and computational methods.

Xenium

Imaging-based (in situ)



 **TRANSCRIPTOME**
Targeted, high-plex panels ($\sim 5,000$ genes)

 **SPATIAL SCALE**
Cellular / subcellular

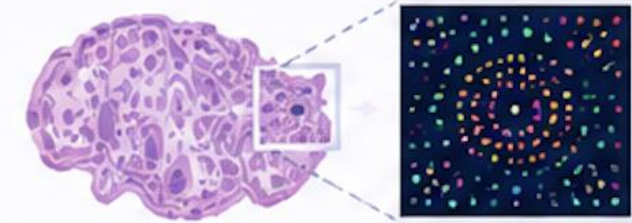
 **TISSUE CONTEXT**
Cellular architecture

 **BEST WHEN...**
Defined genes + cellular localization

 **KEY TRADE-OFF**
Requires panel design; cell segmentation is critical for interpretation.

MERFISH / seqFISH

Imaging-based (in situ)



 **TRANSCRIPTOME**
Targeted / highly multiplexed (100s – 10,000s of genes)

 **SPATIAL SCALE**
Single-molecule / subcellular

 **TISSUE CONTEXT**
Cellular / molecular localization

 **BEST WHEN...**
Very fine molecular detail; cell-cell interactions

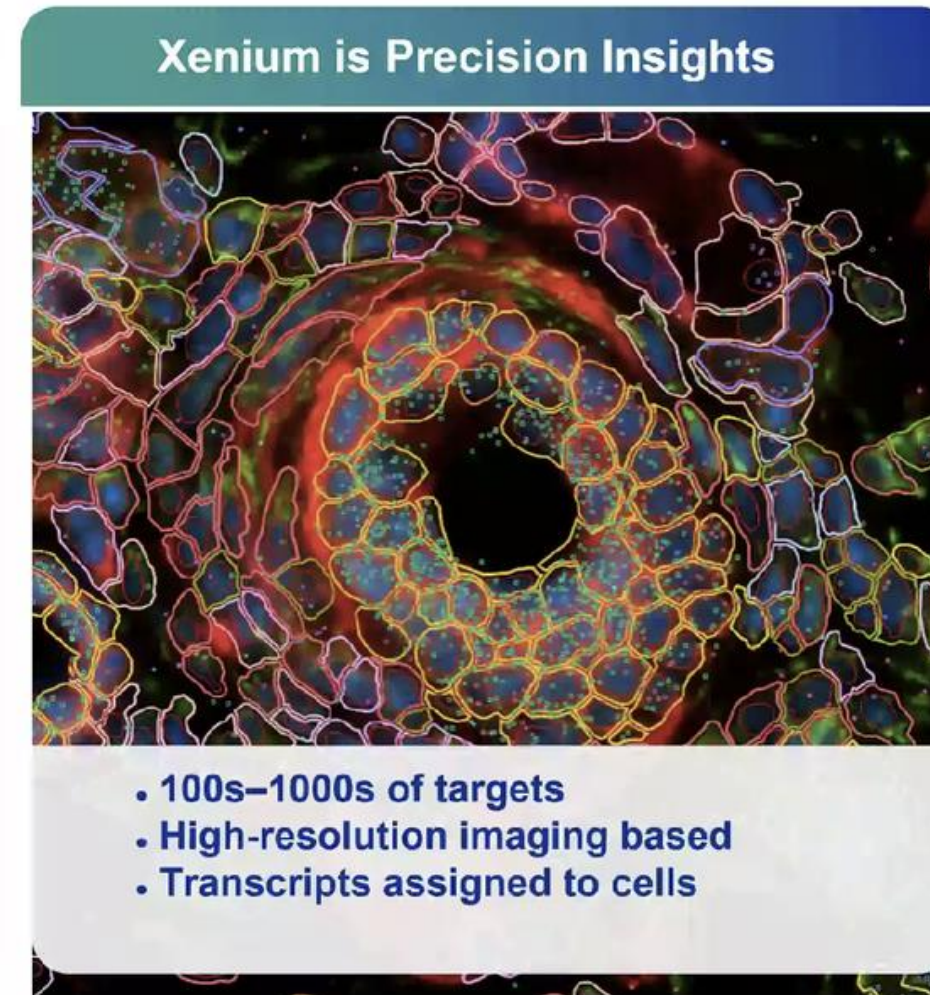
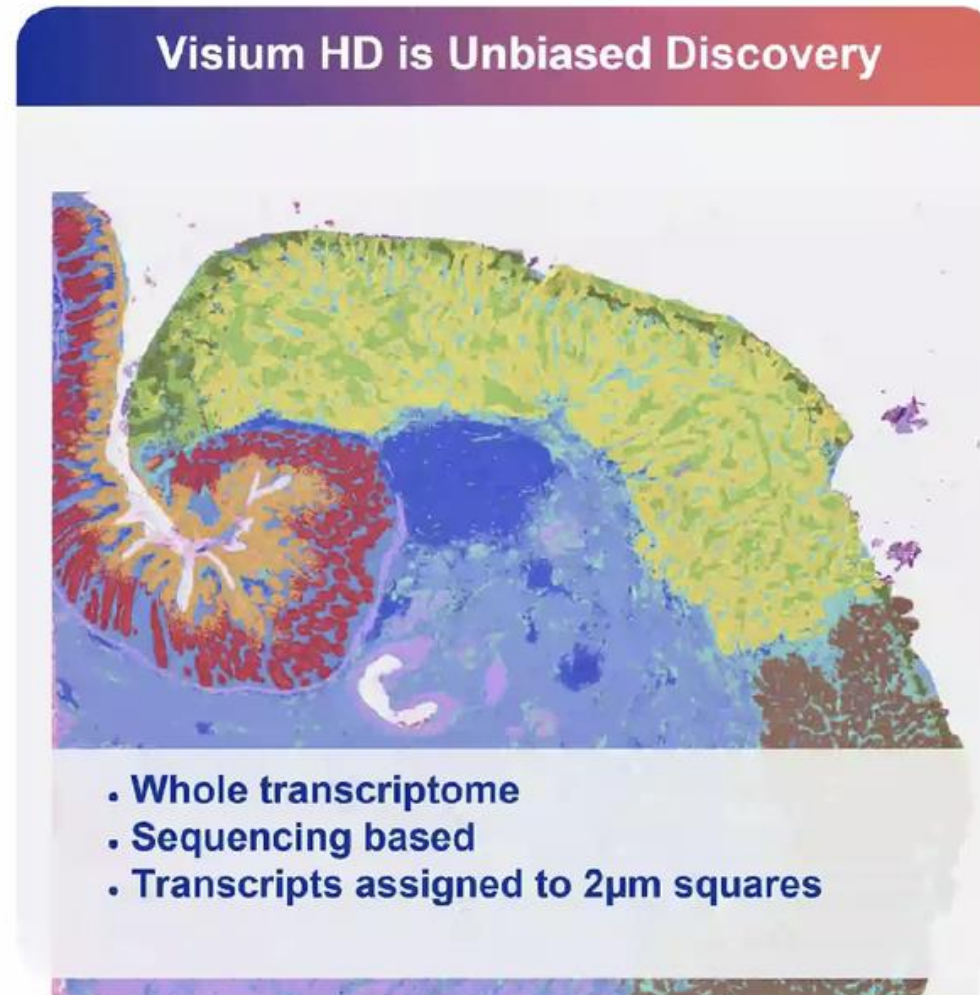
 **KEY TRADE-OFF**
Target selection and higher experimental + computational complexity.



There is no universally “best” spatial assay —
choose the measurement that matches the biological question.

Platform choice can change the quality of data visualized...

...and therefore, the biological interpretation



TIP: One of the best ways to learn a platform's strengths and limitations is to explore public datasets before running your own experiment!

05

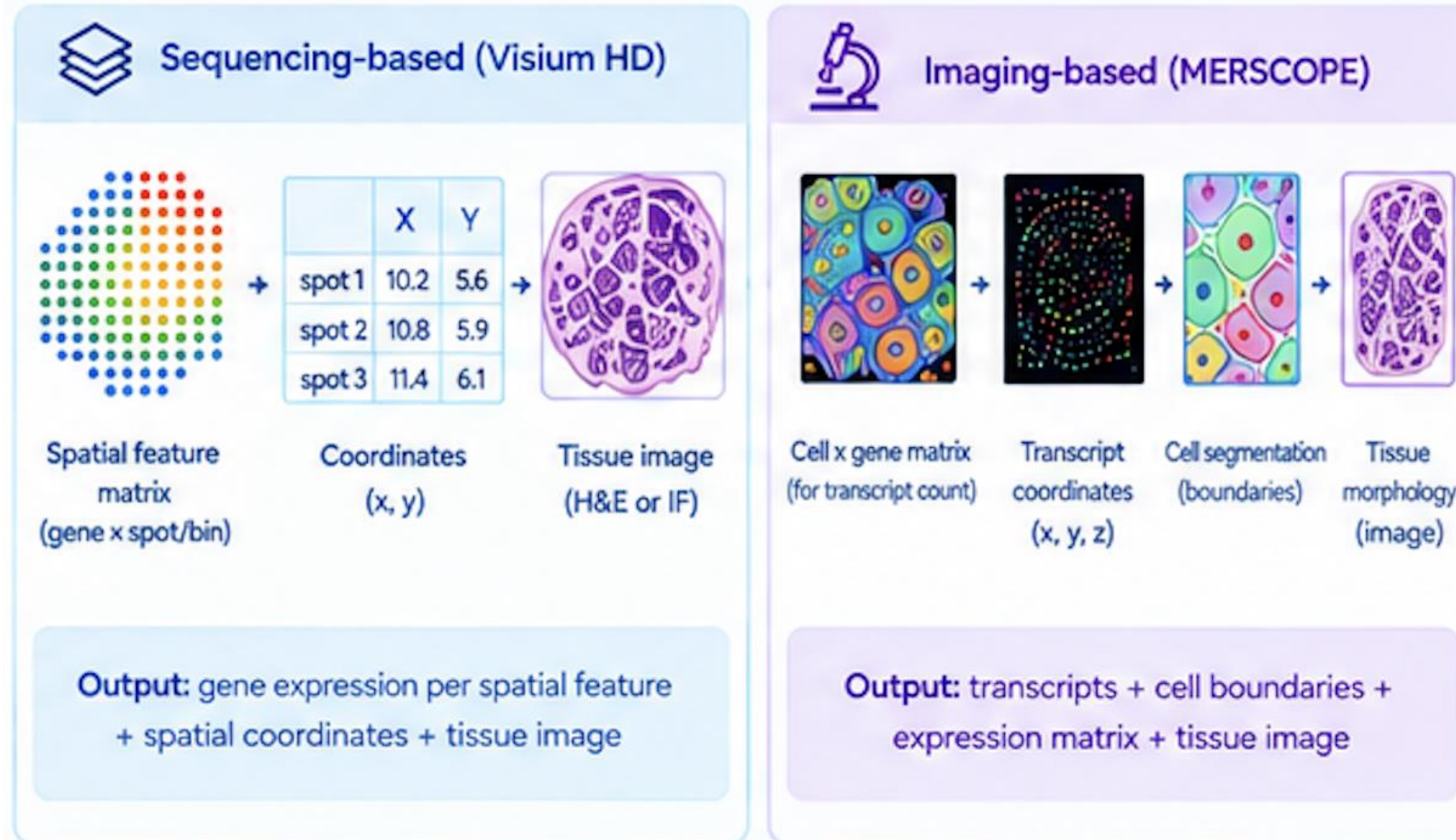
FROM MEASUREMENT TO DATA

The assay determines what the computational object actually is.



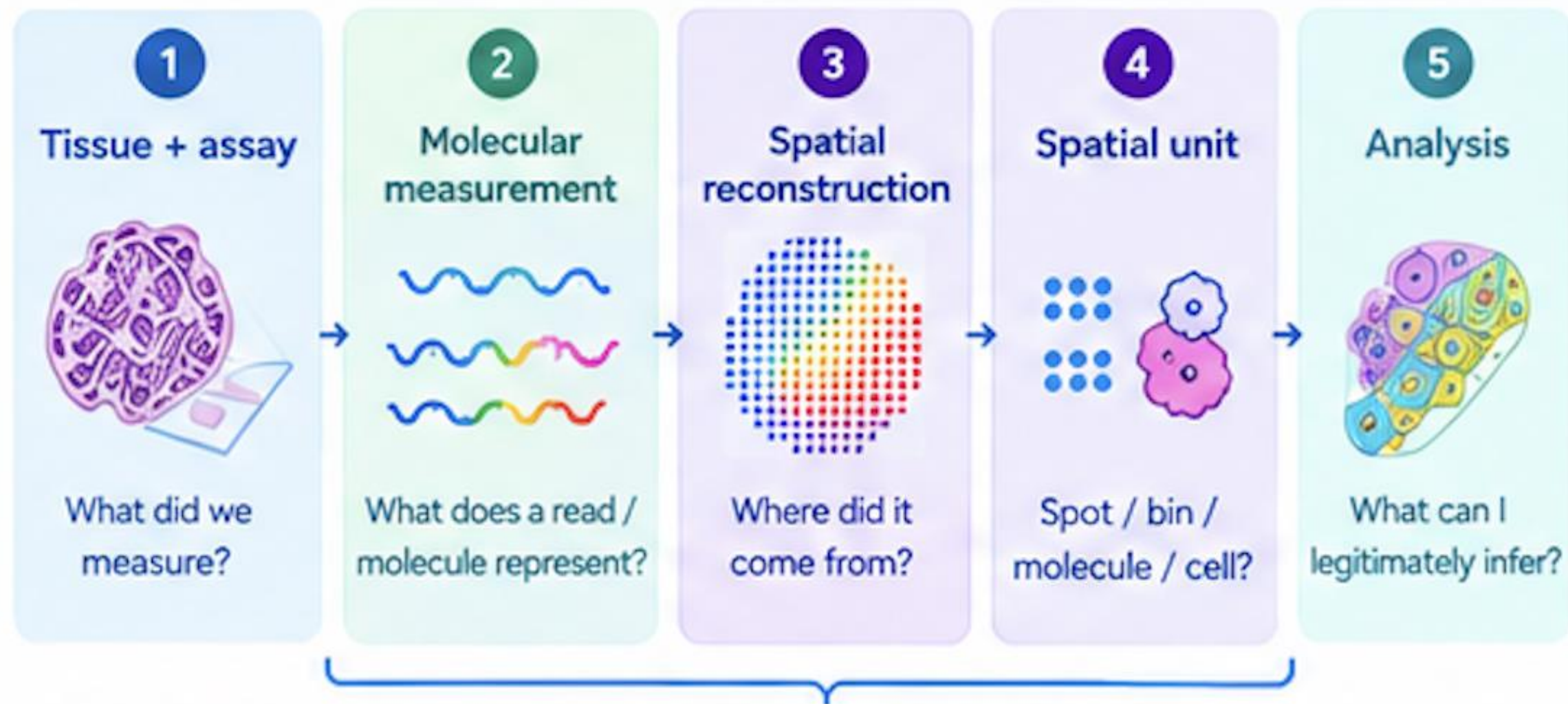
What do we actually get from the experiment?

Different measurement strategies → different data objects.



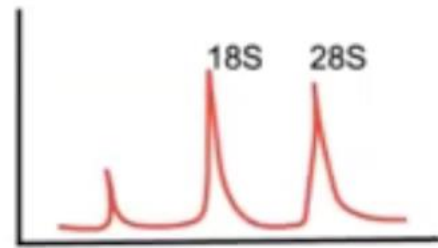
These datasets can both be called spatial transcriptomics, but they are not the same data object. And that difference propagates all the way into the analysis (analyses are not interchangeable)

Your computational analyses start before R or Python



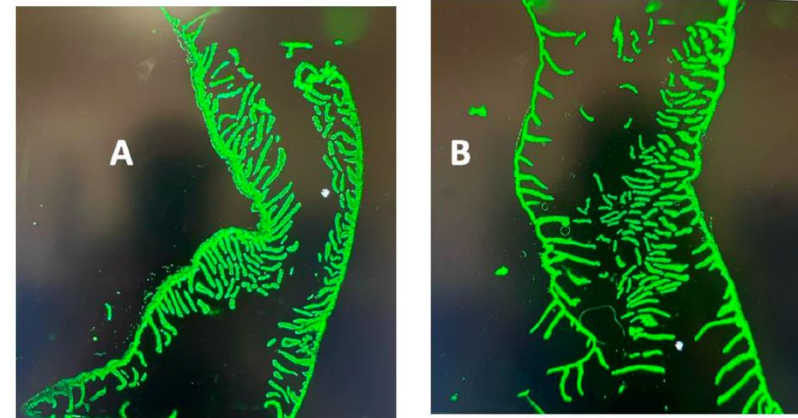
R and Python come later — after key decisions about what was measured and how it was reconstructed

What can go wrong before the data reaches you? (wet bench-side)

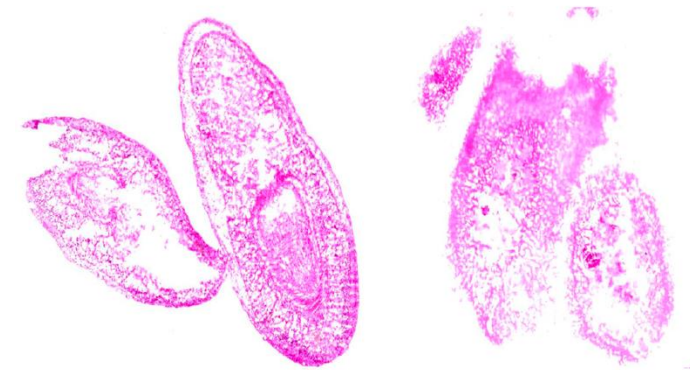


poly(A): RIN > 8-9
probes: DIV >= 200

Poor/Suboptimal RNA



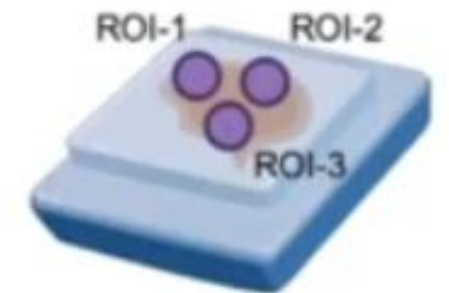
Tissue attachment issues



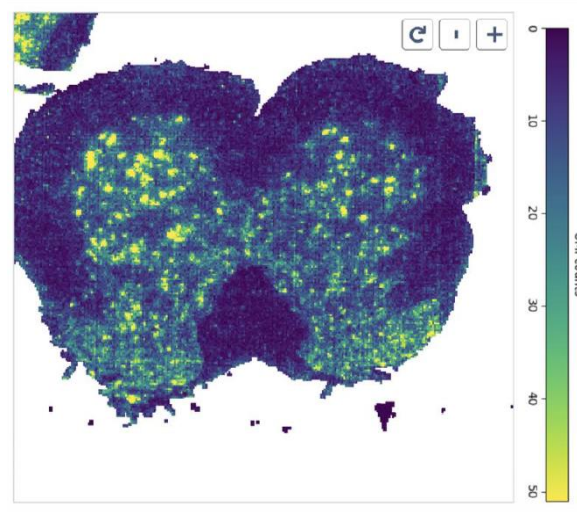
Loss of morphology/lysis/
RNA diffusion



Rough handling



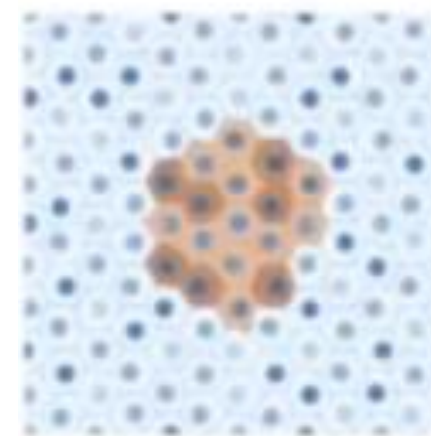
Wrong ROI
selection



Probe binding failure
(Bad permeabilization/
uneven sectioning)



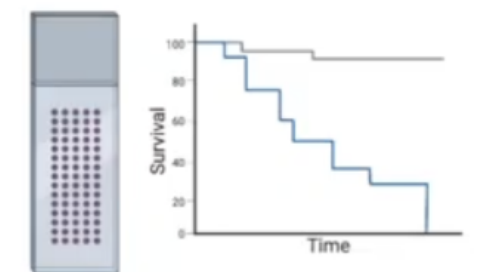
Appropriate assays
not picked or run
(sample – kit – platform
incompatibility)



No key markers/
No reference dataset



No wet bench-computational
coordination



No rep. cohort or
large scale validation

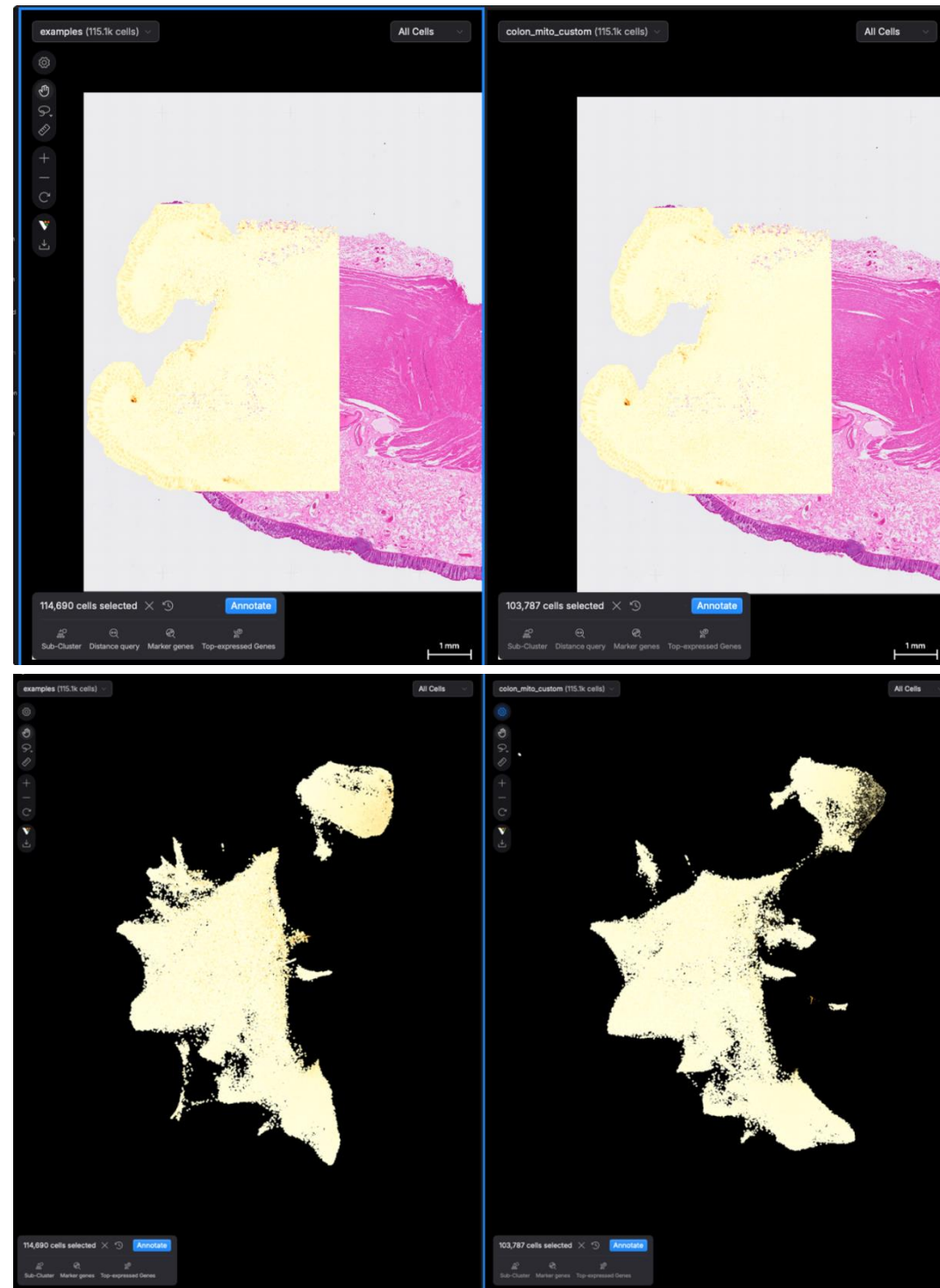
What can go wrong before the data reaches you? (computational-side)

Mistake 1: Over-filtering During Quality Control

Goal is not to remove as many cells as possible, but to remove unreliable measurements while preserving meaningful biological information

Fix: Do not blindly apply QC thresholds from another study w/o considering difference in tissue type, biological context, Sequencing depth, or spatial platform

QC should be contextual, not one-dimensional
E.g. Low UMI $\neq$ Bad data always



Effect of MT% filtering thresholds on a Visium HD colon dataset.

Key Metrics

486,254

Number of 8 μ m binned squares under tissue

1467.1

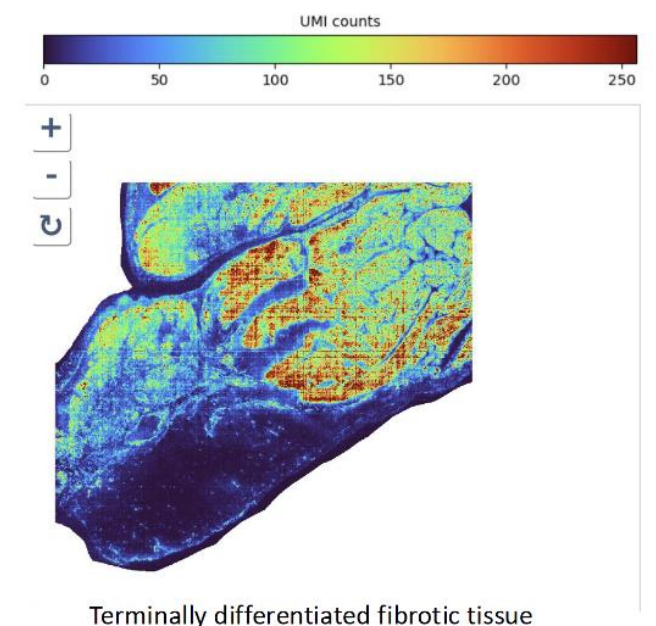
Mean reads per 8 μ m bin

71.1

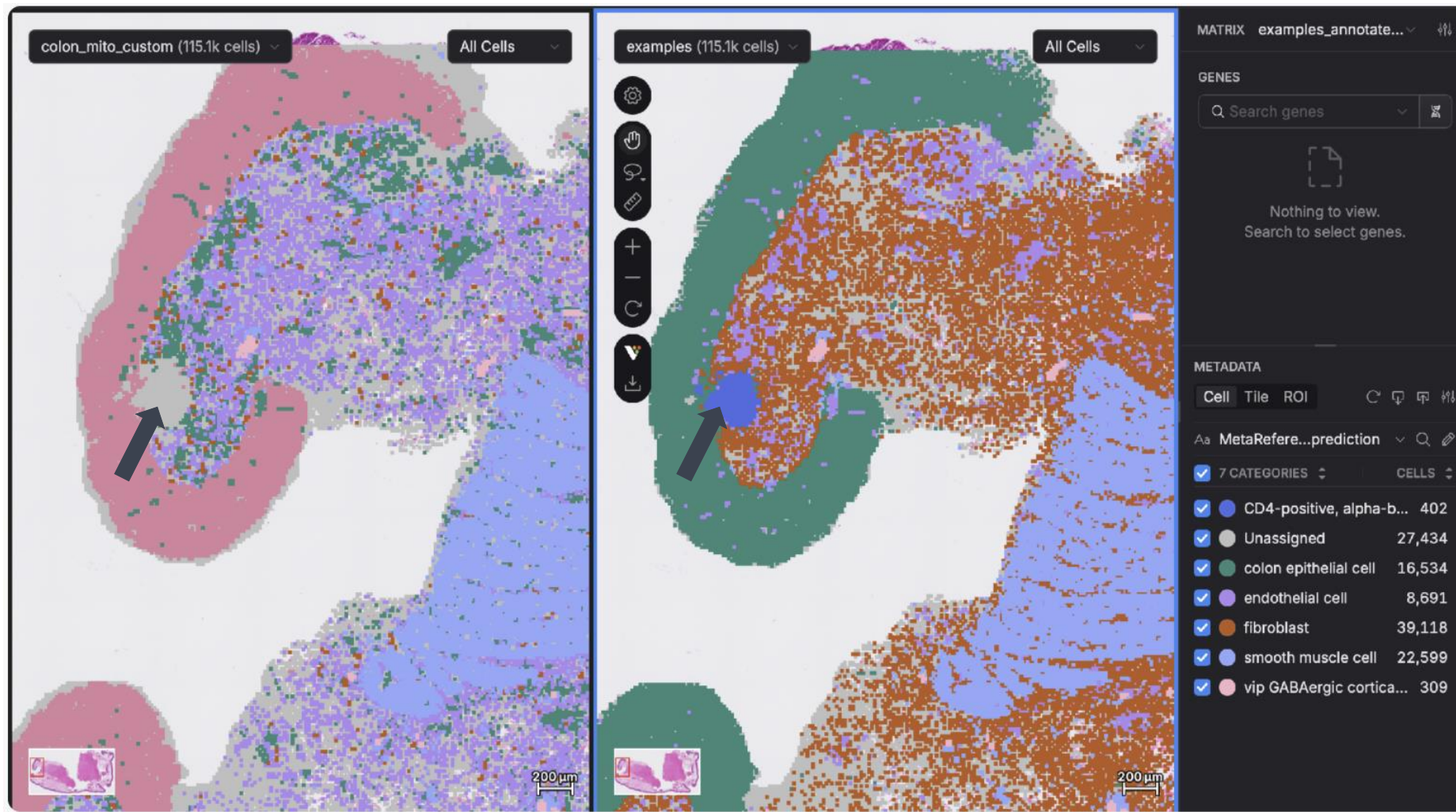
Mean UMIs per 8 μ m bin

17,328

Total genes detected



What can go wrong before the data reaches you? (computational-side)



The same Visium HD dataset as Mistake 1 can yield different annotations depending on QC strategies

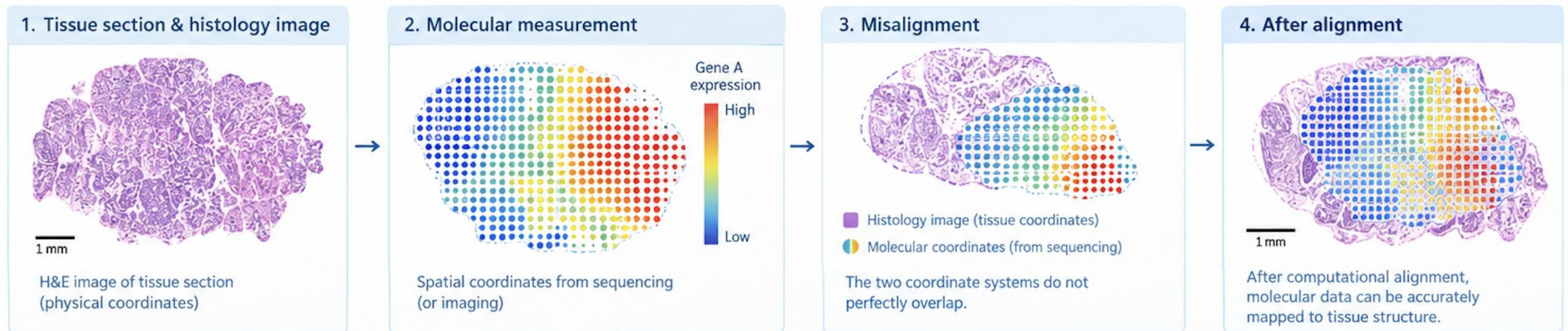
Mistake 2: Trusting a single, unvalidated cell-type annotation

Cell-type annotation is influenced not only by the annotation algorithm itself but also by upstream analytical decisions such as quality control and data preprocessing. A confident label should therefore not be interpreted as definitive biological evidence.

Fix: Compare multiple annotation strategies, such as reference-based and marker-based methods.

Mistake 3: Misalignment: molecular co-ordinates do not match tissue image

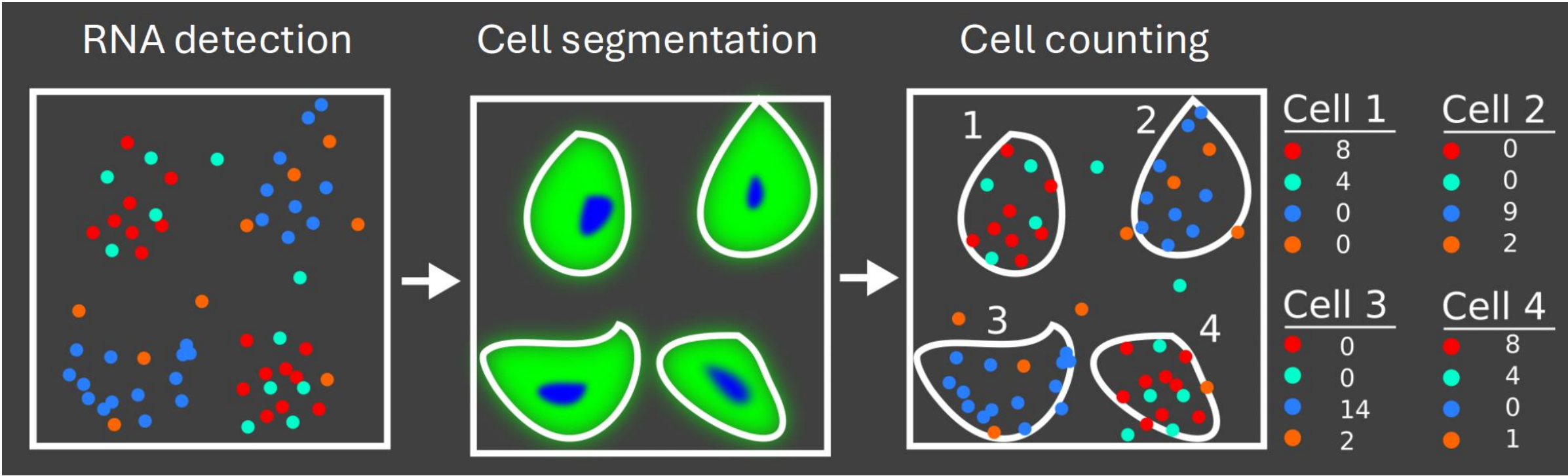
Subtle misalignment (even a few microns) lead to gross misinterpretation



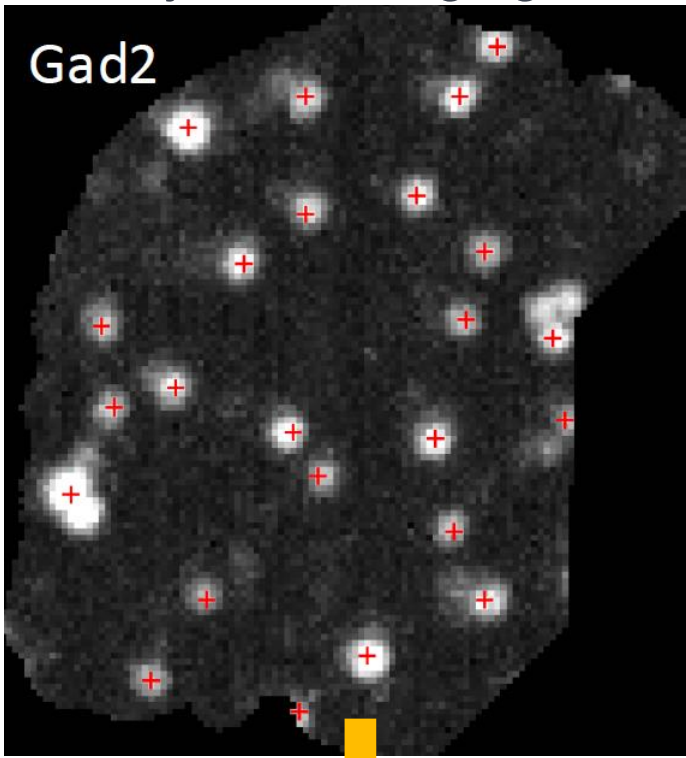
FIX: Never blindly accept default alignment. Inspect and adjust the tissue overlay, then verify against anatomical features or known markers. Misalignment can silently propagate downstream.

Mistake 4: Over or under-segmentation

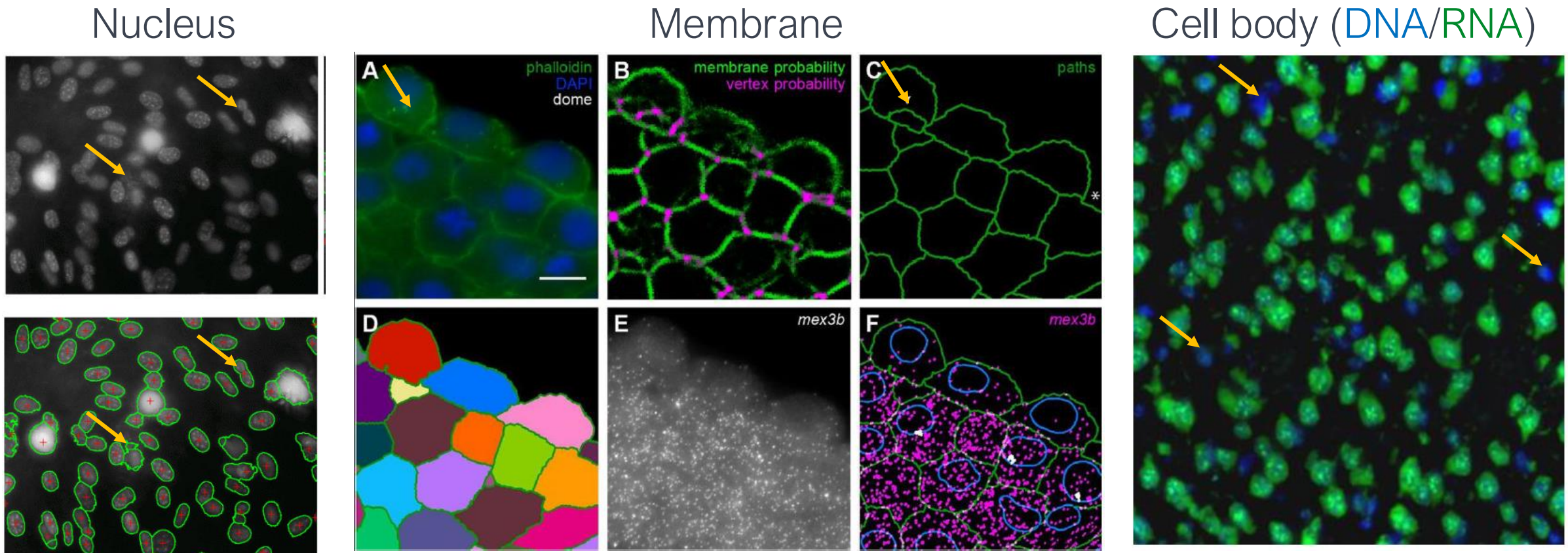
Cell assignment



Terabytes of imaging data



Cell segmentation



	1124	2325	2400	2411	6248	5992	275	2572	1199	5162	532	3607	3253	7173	2757	1229	1204	7797	4653
Hybridization1_Tbr1	13	11	26	12	7	6	14	24	...	57	20	26	8	5	5	0	14	18	5
Hybridization1_Aldoc	38	0	0	5	38	2	4	3	7	10	...	11	10	5	4	10	2	0	5
Hybridization1_Foxj1	0	0	0	1	5	0	3	1	1	1	...	0	0	0	2	1	8	1	2
Hybridization6_Bmp4	1	0	0	0	0	0	0	0	1	0	...	0	0	0	0	0	1	1	0
Hybridization6_Irpr2	4	0	0	1	0	0	1	0	2	1	...	3	0	1	2	3	0	0	0
Hybridization6_Vp	13	1	2	4	30	1	3	2	1	4	...	0	11	2	5	1	7	2	3
Hybridization4_Cnr1	0	0	0	0	65	5	0	0	0	0	...	2	0	9	0	17	0	0	0
Hybridization4_Pip1	16	0	0	0	8	0	0	6	0	0	...	0	0	0	10	1	27	5	1
Hybridization4_Vin	0	0	0	2	4	0	2	1	1	0	...	0	3	1	2	2	0	0	3
Hybridization7_Rorb	4	0	0	1	0	4	0	0	2	3	...	0	27	14	0	0	1	0	1
Hybridization7_Sox10	52	0	1	1	3	3	13	3	19	33	...	1	4	0	10	12	40	15	32
Hybridization7_Ctsp	6	3	8	15	3	3	3	5	2	1	...	6	4	12	14	1	2	0	1
Hybridization11_Syl6	1	16	20	0	0	0	3	21	1	2	...	4	0	1	11	1	3	0	12
Hybridization11_Tbr1	4	13	36	9	2	5	9	12	6	15	...	30	19	30	0	3	2	0	10
Hybridization11_Tmem6	2	0	0	3	1	1	2	2	1	2	...	4	1	3	1	1	0	0	4
Hybridization8_Pdgfra	1	1	2	0	1	0	2	1	20	1	...	1	1	1	2	26	0	0	1
Hybridization8_Serpin1	13	1	2	6	2	4	2	1	10	2	...	0	5	10	8	6	5	6	2
Hybridization8_Pthb	2	0	0	0	6	0	1	1	0	0	...	0	0	1	1	0	0	0	1
Hybridization16_Crbp	2	0	1	0	0	0	0	0	0	3	...	0	0	0	0	0	0	0	0
Hybridization10_Crh	2	0	2	0	3	0	6	1	0	2	...	1	1	1	1	0	0	3	1
Hybridization10_Apln	3	5	2	31	0	2	3	4	8	5	...	0	2	3	1	2	3	1	3
Hybridization9_Lamp5	6	38	51	126	0	1	52	44	51	0	...	4	1	168	5	5	0	1	3
Hybridization9_Lum	1	0	0	3	0	0	0	0	3	6	...	0	0	0	1	1	0	0	0
Hybridization9_Atn	19	1	1	1	2	2	2	5	3	23	...	0	1	10	3	8	14	8	11
Hybridization12_Kcnp	1	25	50	14	6	3	20	14	7	0	...	25	23	64	0	2	0	0	3
Hybridization12_Slc32a1	2	1	2	2	22	0	1	0	2	0	...	0	1	1	0	0	4	0	0
Hybridization12_Vin	2	2	0	1	2	0	0	0	0	2	...	0	1	0	0	2	0	0	1
Hybridization5_Acta2	3	1	1	1	1	0	1	4	0	2	...	0	0	2	0	7	6	0	4
Hybridization5_Cpne5	0	4	1	1	1	0	2	9	2	0	...	3	0	10	0	3	0	0	3
Hybridization5_Klf6	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	2	0	1
Hybridization3_Mtga9	6	0	1	2	0	2	2	3	2	13	...	2	2	2	1	7	4	0	2
Hybridization3_Mrc1	14	2	2	3	2	0	0	6	1	19	...	2	2	26	4	9	0	1	6
Hybridization3_Hoxb	10	0	3	6	1	0	3	4	1	3	...	9	2	4	3	6	2	0	1
Hybridization2_Gad2	7	4	3	5	65	1	7	1	2	12	...	2	6	9	2	2	1	0	3
Hybridization2_Fit1	0	0	0	0	0	0	0	0	0	0	...	0	3	0	0	0	0	0	0
Hybridization2_Cfap	57	0	1	0	1	0	3	3	0	32	...	3	1	4	5	0	0	1	2
Hybridization13_Cnr1	1	1	2	14	56	3	6	7	5	3	...	3	3	25	1	2	3	0	0
Hybridization13_Ttr	2	0	0	1	1	0	1	0	1	13	...	0	2	0	3	3	1	1	3
Hybridization13_Pip1	10	5	3	0	7	0	0	8	5	33	...	3	1	2	36	0	4	7	39

Panagiotakis et al., 2018

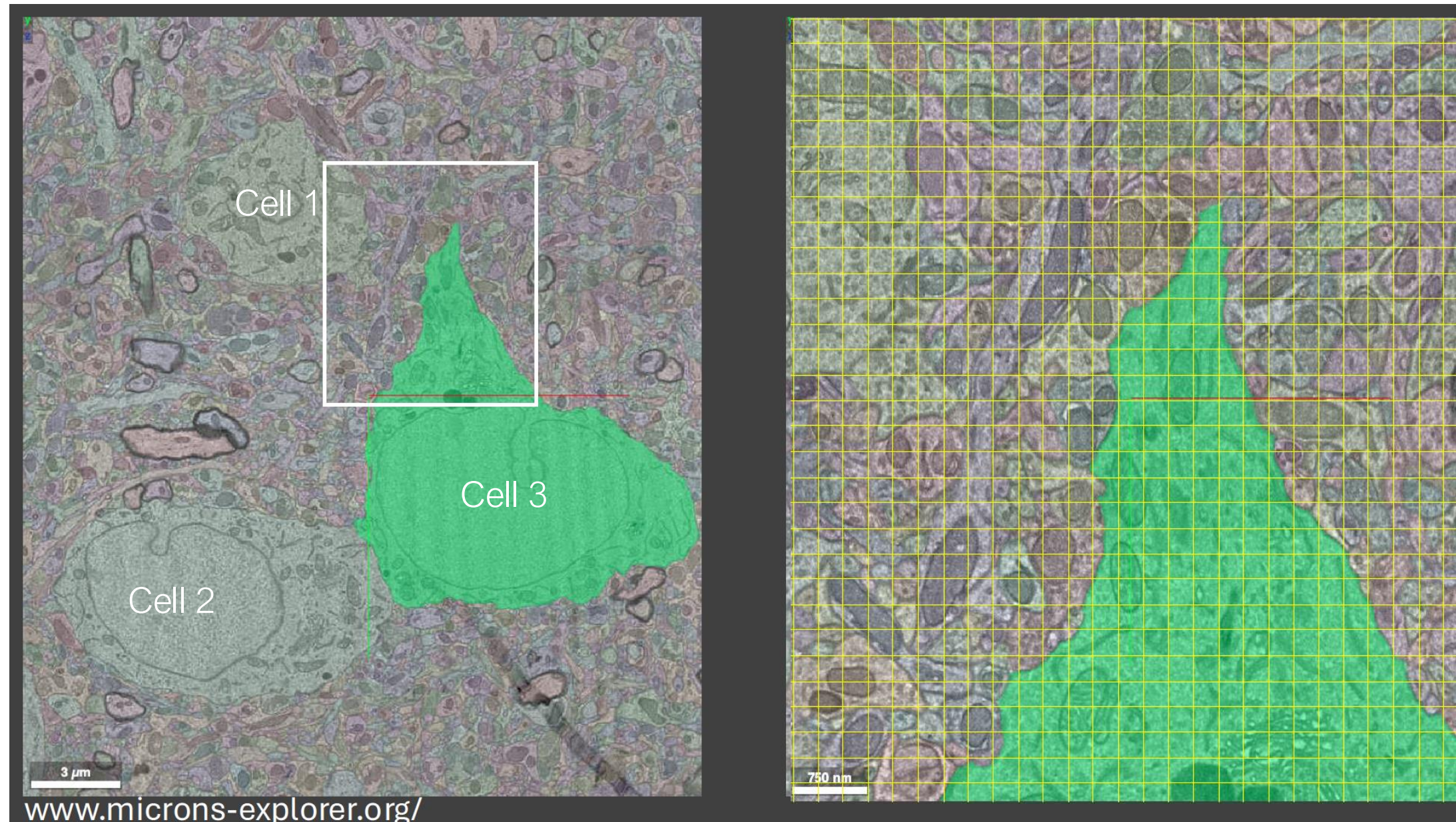
Stapl et al., 2016

Codeluppi et al., 2018

Gb/Mb of gene x cell

Fix: Overlay segmentation boundaries on the raw morphology or DAPI image

Electron microscopy image of mouse brain



Don't treat Segmentation as a black box

Limitations:

- Limited counter stains
- Unclear ground truth
- Resolution

Progress:

- More counter stains
- Better algorithms
- Segmentation-free approaches

Ultimately, a computational cell should correspond as closely as possible to a biological cell, not simply an object generated by a segmentation algorithm

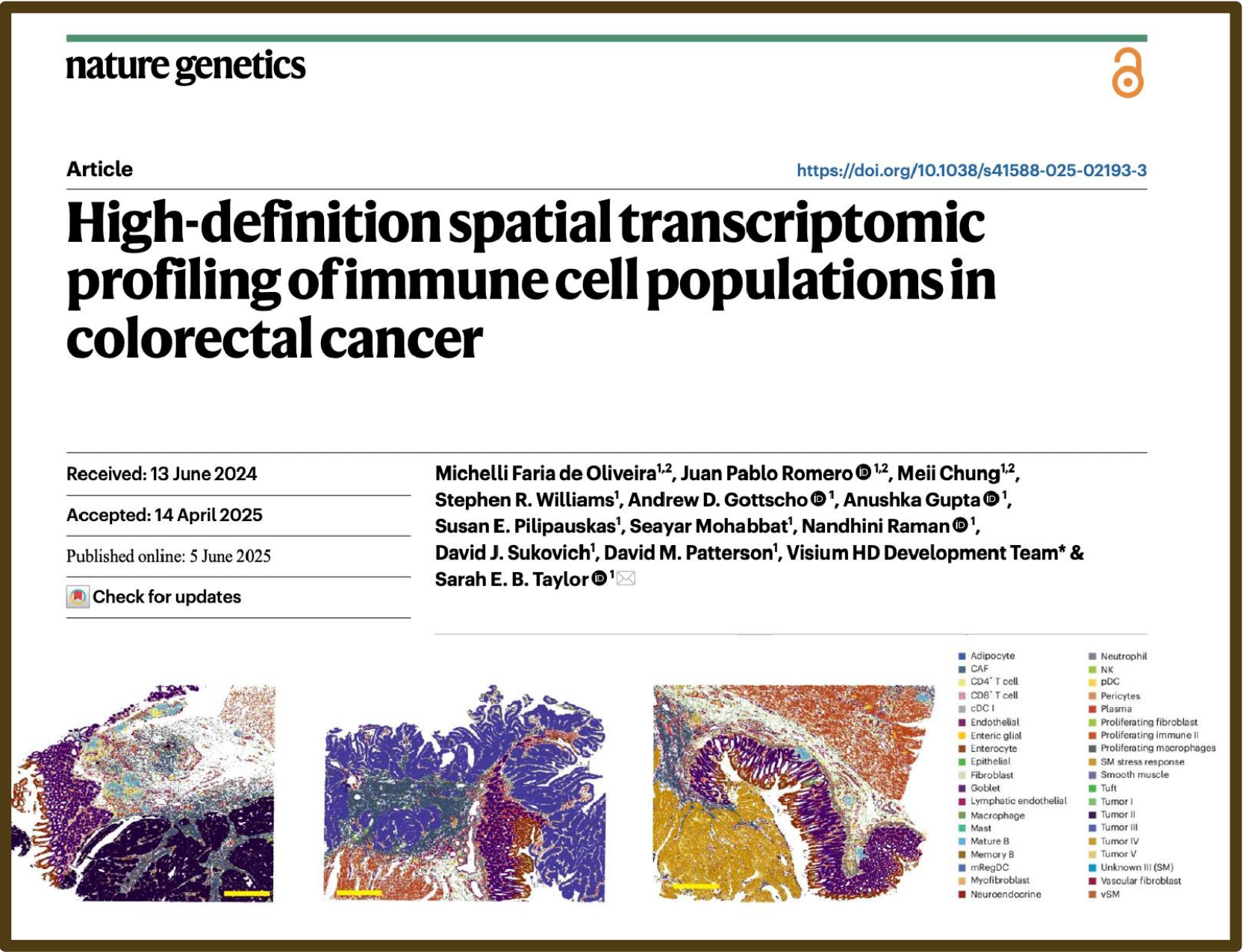
FROM DATA TO BIOLOGY

Now the same spatial dataset becomes a sequence of biological questions.



This is the dataset you will actually learn to analyze

de Oliveira et al., 2025 – the dataset behind this workshop



THE QUESTION

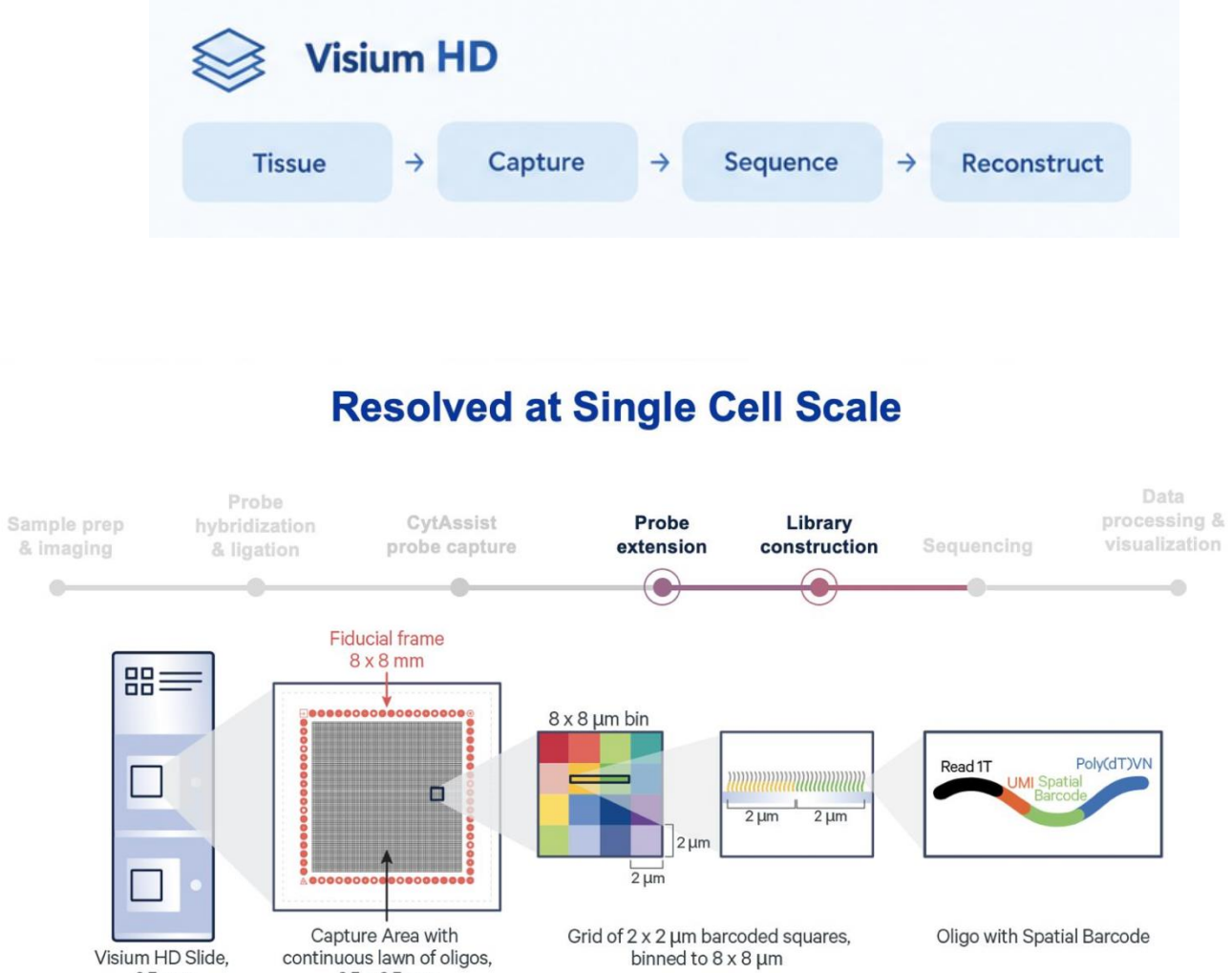
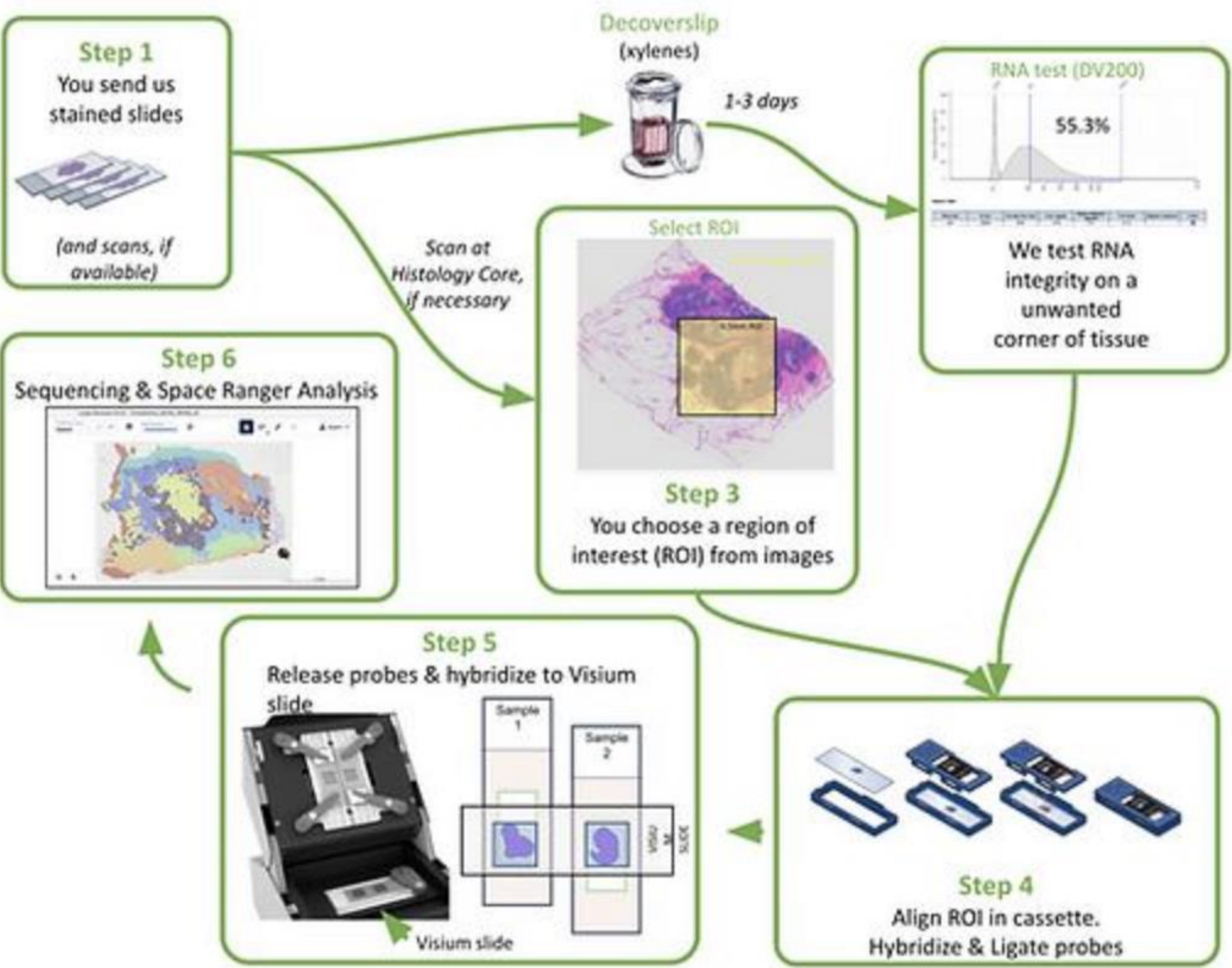
Where are distinct immune and tumor states located within the CRC tissue architecture?

DISCOVER → MAP → INTERPRET

- Single-cell data provide a reference repertoire.
- Visium HD preserves tissue context at high-density spatial sampling.
- Spatial analysis asks how those states are organized.

This is the dataset you will actually learn to analyze

de Oliveira et al., 2025 – the dataset behind this workshop



This is the dataset you will actually learn to analyze

de Oliveira et al., 2025 – the dataset behind this workshop

Workshop: load → QC → filter → single-cell analysis → spatial analysis

QC

Can I trust the measurement?

Clustering

What molecular states are present?

Deconvolution / reference mapping

What cell populations contribute here?

Spatially variable genes

Which genes change with location?

Neighborhoods

Which populations tend to occur near each other?

Cell-cell communication

What molecular interactions might be occurring?

Each analysis adds a different biological question — not just another plot.

Key Take-Aways

- Spatial transcriptomics is a family of measurement strategies, preserving info that dissociation destroys
- Sequencing and imaging-based platforms preserve spatial information differently
- Start with the biological question, not the platform
- A spatial measurement unit is not automatically a biological measurement unit (i.e. a cell)
- The measurement determines the data; the data determines what you can infer
- A spatial map is evidence — not biological truth — until validated

Future of Spatial:

Technological Development and computational advances (3D reconstruction, AI integration)

Clinical (Pathology) integration & Translation/Precision Medicine

From molecules to maps

Spatial measurements let us ask
not only “what is here?” — but “where, and in what
context?”

The measurement determines the map.
The map expands the biological question.